

Heat Stress Induces Phage Tolerance in Bacteria

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 Copyright information

Reviewed Preprint

v2 • June 19, 2025

Revised by authors

Reviewed Preprint

v1 • April 2, 2025

eLife Assessment

This **important** study analyzes the effect of heat treatment on phage-bacterial interactions and **convincingly** shows that prior heat exposure alters the bacterial cell envelope, enhancing persistence and bacterial survival when exposed to lytic phages. The study will interest researchers working on antibiotic resistance, tolerance, and phage therapy.

<https://doi.org/10.7554/eLife.105703.2.sa3>

Abstract

Antibiotic resistance and tolerance present significant challenges in global healthcare, necessitating alternative strategies such as phage therapy. However, the rapid emergence of phage-resistant mutants poses a potential risk. Here, we investigated bacterial persistence against phages, characterized by heterogeneous survival, analogous to antibiotic persistence. We found that heat treatment enhanced persistence and increased bacterial survival under phage exposure, subsequently promoting the evolution of phage resistance. Further experiments demonstrated that heat stress leads to a reduction in envelope components, thereby inhibiting phage DNA injection. Additionally, this heat-induced reduction resulted in systematic alterations in envelope stress responses, rendering bacteria tolerant to the antibiotic polymyxin while making them hypersensitive to pH changes and immune clearance. Our findings provide novel insights into bacteria-phage interactions and highlight potential challenges in implementing phage therapy in clinical settings.

Importance

This study identifies bacterial persistence against phages, showing that heat treatment enhances this persistence and promotes phage resistance evolution by altering envelope components, which affects both phage susceptibility and antibiotic tolerance, highlighting challenges for clinical phage therapy.

Introduction

Antibiotic resistance poses a significant threat to global public health, emphasizing the urgent need for innovative strategies to combat bacterial infections (Luong et al., 2020 [↗](#)). Phage therapy has emerged as a promising alternative, yet the rapid emergence of phage-resistant bacterial mutants challenges its effectiveness (Oechslin, 2018 [↗](#); Salmond & Fineran, 2015 [↗](#); Strathdee et al., 2023 [↗](#)).

Beyond resistance, antibiotic tolerance presents a critical concern, enabling bacteria to survive high concentrations of antibiotics (Balaban et al., 2019 [↗](#); Balaban et al., 2004 [↗](#); Brauner et al., 2016 [↗](#)), which can further promote the evolution of resistance (Levin-Reisman et al., 2019 [↗](#); Levin-Reisman et al., 2017 [↗](#); Liu et al., 2020 [↗](#)). This raises an important question: Could analogous persistence and tolerance arise in response to phage infections, thereby contributing to the development of phage resistance?

Results

Bacterial Persistence Against Phage Infection

To address this question, we investigated the killing dynamics of *Klebsiella pneumoniae* under treatment with the lytic phage Kp11 (GenBank: ON148528.1, **Supplementary Fig.S1** [↗](#)) (Huang et al., 2024 [↗](#)). In analogy with the concept of antibiotic persistence, the majority of bacteria were rapidly eliminated following lytic phage treatment, while a small fraction ($\sim 10^{-5}$ to 10^{-6}) survived, resulting in a bi-phasic killing curve (**Fig.1b** [↗](#)). However, the bacteria survived likely comprised both persisters and resistant mutants. To quantitatively distinguish between resisters and persisters, colonies of surviving bacteria post-phage treatment were individually plated on media with and without phage. The fraction of persisters was estimated by subtracting the number of resisters from the total colony-forming units (CFU) (**Fig.1a, 1b** [↗](#)). Subsequent killing experiments on the descendants of persisters revealed a comparable survival fraction to the ancestral population, suggesting that this survival was not attributable to genetic mutation but rather to phenotypic heterogeneity, i.e., phage persistence (**Fig.1c** [↗](#)).

Canonical antibiotic persistence is characterized by an extended lag time or slow growth rate (Balaban et al., 2004 [↗](#); Brauner et al., 2016 [↗](#)). However, previous studies showed that the high antibiotic persistence *Escherichia coli* strain *hipA7* did not exhibit enhanced survival when exposed to lytic phages compared to its sensitive ancestor (Pearl et al., 2008 [↗](#)). As there currently lacked a well-defined high persistence variant of *K. pneumoniae*, we turned to the *E. coli* strain *KLY* and its high-persistence derivative *metG** and challenged these two well-studied strains with T4 phage, which was known for its lytic activity against *E. coli*. The observed similar survival rates suggested that increased lag time does not confer protection against phage infection (**Fig.1d** [↗](#) and **Supplementary Fig.S2** [↗](#)).

Phage tolerance induced by heat stress

Given that antibiotic persistence can be triggered by stresses such as starvation (Betts et al., 2002 [↗](#); Nguyen et al., 2011 [↗](#)), we explored whether phage persistence might similarly be affected or not. *K. pneumoniae* was subjected to various stress conditions prior to phage treatment. Remarkably, pre-treatment at elevated temperatures significantly reduced *K. pneumoniae*'s susceptibility to lytic phage infection (**Fig.2a** [↗](#)). Killing assays revealed that heat-treated bacteria exhibited a reduced killing rate following phage exposure (**Fig.2b** [↗](#)), with effects being both time-

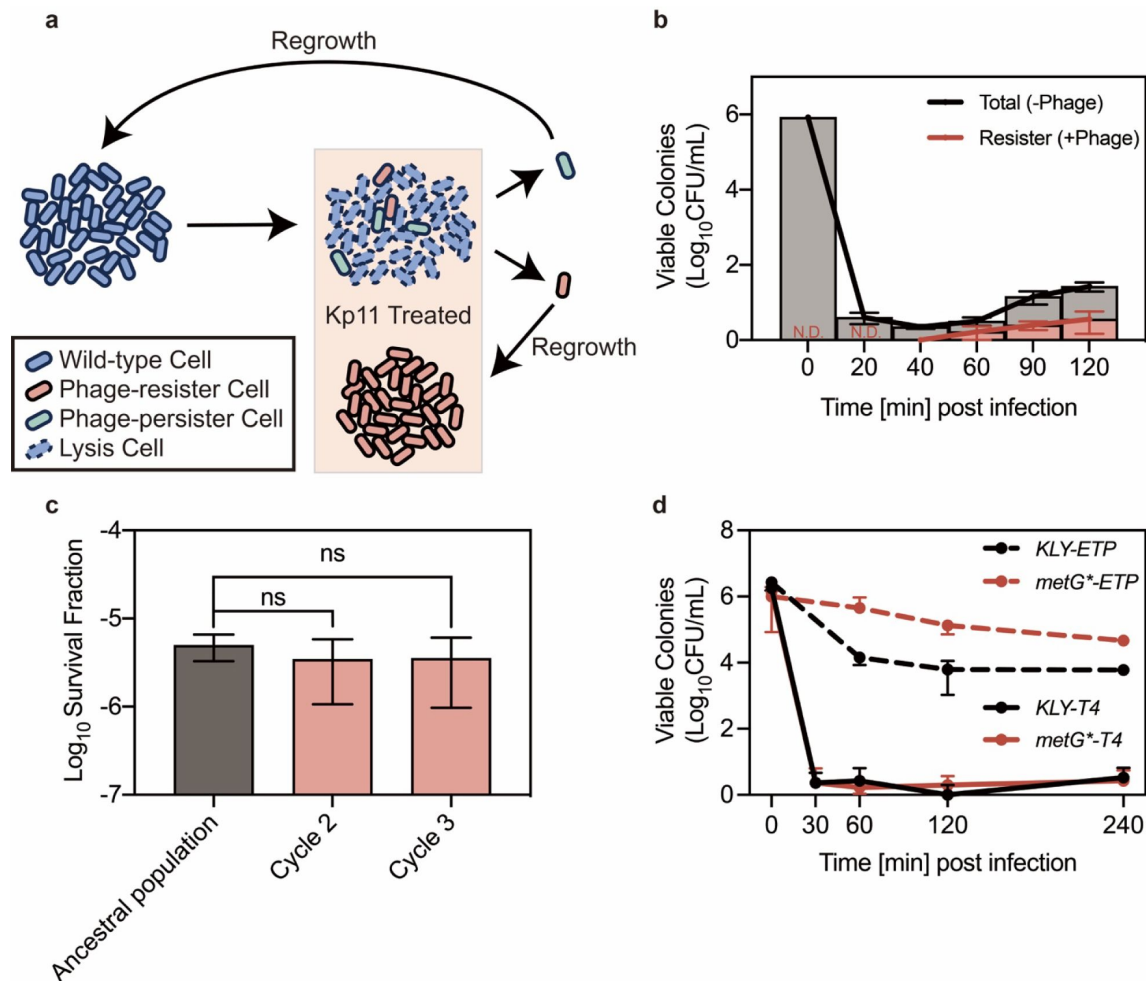


Fig.1

Bacterial persistence against phage infection.

a. Schematic illustration of the experiments demonstrating the existence of phage persisters. **b.** Viable colonies of *K. pneumoniae* ATCC 43816 at indicated time points post-infection with phage Kp11 (multiplicity of infection, MOI ~ 100), grown on both phage-containing (+phage) and phage-free (-phage) plates. Curves represent the mean $\pm$ standard deviation (s.d.) of three biological replicates. **c.** Survival fraction after 40 minutes of phage Kp11 treatment for the ancestral strain ATCC 43816 and descendants of persisters. Data are presented as mean $\pm$ s.d. of three biological replicates; ns indicates not significant ($P = 0.6131$ and 0.6415). **d.** Viable colonies of *KLY* and its high-persistence derivative strain *metG** that survived treatment with the antibiotic ertapenem ($5\mu\text{g/mL}$, $20 \times \text{MIC}$) and phage T4 infection (MOI ~ 100). Curves show the mean $\pm$ s.d. of three biological replicates.

and temperature-dependent (**Supplementary Fig.S3**). Moreover, infection curves indicated that heat-treated bacteria relapsed more frequently, suggesting a higher probability of resistance mutant emergence (**Fig.2c**).

To determine whether heat treatment directly conferred increased phage resistance, we quantified the fraction of phage-resistant mutants before and after heat treatment. Survivors from the phage killing assay were collected and plated on both phage-containing and phage-free plates, with only resistant colonies detectable on the phage-containing plates. Our analysis revealed a negligible presence of resistant mutants in both groups (**Fig. 2d**). Consistent with this, mutation rates estimated using Delbrück-Luria tests (Luria & Delbrück, 1943) confirmed no significant difference (**Fig.2e**). Furthermore, although the survival rate of heat-treated bacteria substantially increased after 2 hours of phage infection, the proportion of phage-resistant mutants remained comparatively small (**Fig.2d**). These findings indicated that the enhanced defense against phage infection observed following heat treatment was not due to an increase in phage-resistant mutants, but rather reflected a phenotypic boost in survival analogous to antibiotic persistence (Brauner et al., 2016). Therefore, we propose that the observed increase in resistance frequency is due to the cumulative effect of a net increase in survival, similar to how antibiotic tolerance promotes resistance.

Inspired by this concept, we conducted experiments using a very small bacterial inoculum ($\sim 10^3$) to minimize the presence of preexisting resistant mutants, ensuring that any observed resistance arose during phage infection. To maximize phage-bacteria interactions, we used a high multiplicity of infection (MOI ≥ 1) and a small culture volume (100 μ L). After 24 hours of incubation with or without heat treatment and phages, untreated bacteria were completely eradicated, resulting in clear wells (**Fig. 2f**). In contrast, turbidity - defined as an OD₆₀₀ > 0.9 - was observed in wells containing heat-treated bacteria co-cultured with lytic phages, indicating the emergence and growth of resistant mutants during treatment (**Fig. 2f**).

Microscopic observations further revealed that heat-treated bacteria could divide under phage pressure, while untreated bacteria were lysed by phages before dividing (**Fig.2g** and **Supplementary Video 1** and **2**). This indicates that the tolerance induced by heat treatment facilitates the emergence and evolution of phage resistance, as heat-treated bacteria underwent more division events under phage pressure, ultimately increasing the likelihood of developing phage resistance (Levin-Reisman et al., 2017; Liu et al., 2020).

Heat-Induced Phage Tolerance challenges phage therapy

We next assessed the infection of *K. pneumoniae* by phages Kp7, Kp9, Kp10, and Kp11 (Huang et al., 2024), as well as *E. coli* by phages T1 and T4, following pre-heat treatment. Our results revealed a consistent tolerance response across these experiments (**Fig.3a** and **Supplementary Fig.S4a, S4b**). This observation was further validated using clinical *K. pneumoniae* strains K359 and K381, both of which exhibited a marked reduction in phage sensitivity post-heat treatment (**Fig.3b** and **Supplementary Fig.S4c**).

Phage therapy often involves combining multiple phages to minimize the risk of resistance evolution (Chan & Abedon, 2012; Leon & Bastias, 2015). In light of this, we extended our investigation to phage combinations and found that heat-treated bacteria exhibited reduced sensitivity even to combined phage treatments (**Fig.3d**). Notably, we identified a Kp7-resistant strain, R^{Kp7} , which remained susceptible to Kp11 (**Supplementary Fig.S5a**) and was also sensitive to the combination of Kp7 and Kp11 (**Supplementary Fig.S5b**). This approved the effectiveness of combination therapy against spontaneous resistance-conferring mutations to a single phage (Leon & Bastias, 2015). Intriguingly, however, post heat-treatment, the combination

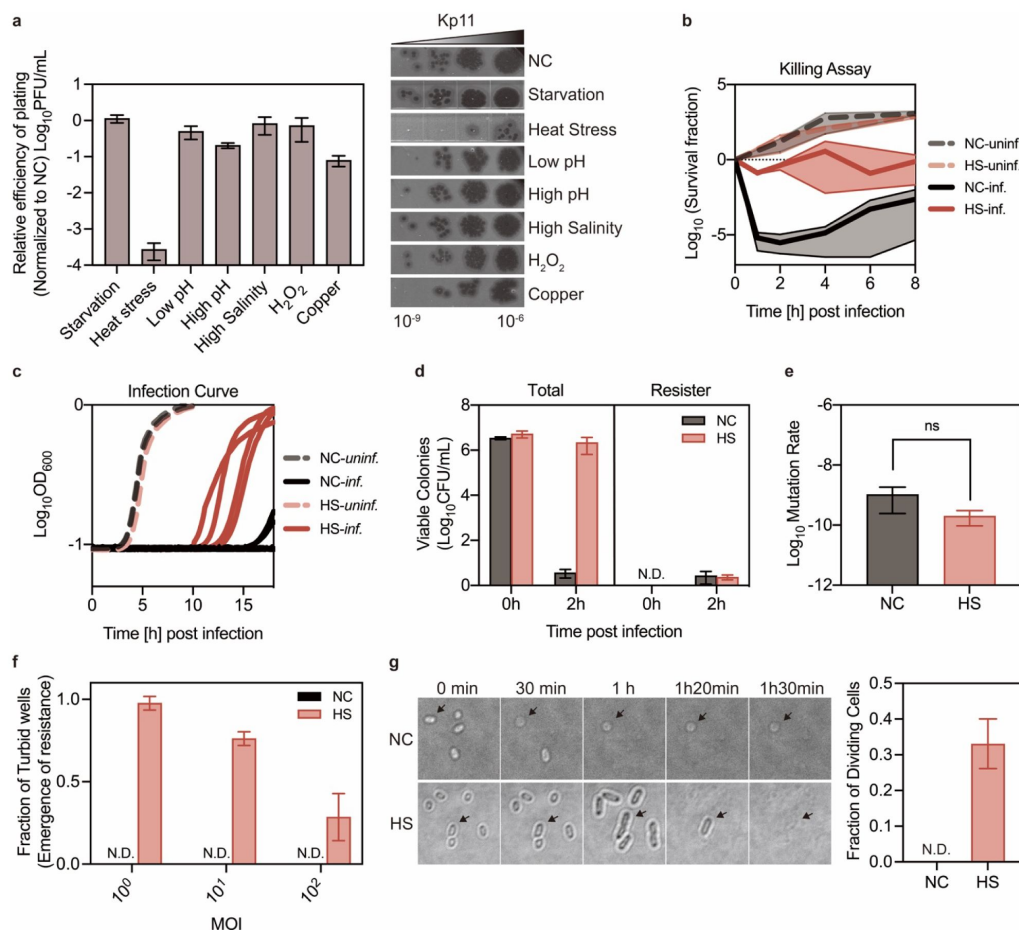


Fig.2

Phage tolerance induced by heat stress in *Klebsiella pneumoniae*.

a. Plaque-forming units of the infection drop assay in *K. pneumoniae* ATCC 43816 following various pre-treatments. The morphotypes of bacteria corresponding to each stress pretreatment are indicated: starvation for 36 hours in 0.9% NaCl; heat stress for 1.5 hours at 50°C; and treatment for 1 hour with low pH (pH 3.2), high pH (pH 10), high salinity (5% NaCl), oxidative damage (10 mM H₂O₂), and copper (5 mM CuSO₄). Data are presented as mean ± s.d. of three biological replicates. NC: Negative Control, bacteria without high-temperature treatment prior to phage infection. **b.** Survival of *K. pneumoniae* ATCC 43816 pre-treated with heat and subsequently infected with phage Kp11 (MOI ~ 100). Curves represent the mean ± s.d. of five biological replicates, with the solid line indicating the mean and the shaded area representing the s.d. of the mean. HS: Heat Stress, bacteria subjected to high-temperature treatment at 50 °C for 90 min prior to phage infection. **c.** Growth dynamics of liquid cultures of *K. pneumoniae* ATCC 43816, with and without heat treatment, infected with phage Kp11 (MOI ~ 100) at 37°C. Curves show the mean of four independent replicates. **d.** Bacteria from the killing assay (panel b, time 0 and 2 h) were collected and plated on plates with and without phage Kp11 (10⁸ PFU). No phage-resistant colonies were detected at time 0 (N.D.: Non-detectable), and similar numbers of phage-resistant colonies were observed at time 2 hours in both populations. Data are presented as mean ± s.d. of three independent replicates. **e.** Mutation rates of *K. pneumoniae* ATCC 43816 with and without heat treatment are shown. Data are presented as mean ± s.d. from five independent replicates and 14 plates for each replicate; ns indicates non-significant ($P = 0.3194$). **f.** Bacteria (~ 10³ CFU) with and without heat treatment were cultured with phage Kp11 at various MOIs of 1, 10 and 100 for 24 hours, with the fraction of turbid wells shown indicating resistance emergence. Data are presented as mean ± s.d. of three independent replicates and 14 wells for each replicate and no turbid wells were detected in negative control groups which without high temperature treatment. **g.** Microscopic analysis of *K. pneumoniae* ATCC 43816 in response to phage Kp11 infection: (Upper Panel) Untreated bacteria were lysed by phages before cell division could occur, whereas (Lower Panel) heat-treated bacteria were able to undergo cell division despite exposure to the lytic phage. Phage Kp11 was added at time 0; scale bar is 2µm; Black arrows indicate example cells undergoing lysis or division. Quantitative analysis further demonstrated a significantly higher fraction of dividing cells in the heat-treated group under phage pressure compared to untreated bacteria.

had a much higher propensity to fail due to the evolution of resistance to both phages (**Fig.3d** [↗](#)). These findings highlight the necessity of considering phage tolerance as a crucial factor in the design of effective phage therapies.

Reduced Phage DNA Entry and Altered Envelope Stress Response in Heat-Treated Bacteria

To investigate the mechanisms underlying phage tolerance in heat-treated bacteria, we first focused on the early stages of phage infection: adsorption and DNA injection. Our analysis revealed that heat treatment did not alter the phage adsorption rate (**Fig.4a** [↗](#)). Additionally, the production and structure of the bacterial capsule, the primary receptor for phage Kp11, remained unchanged (**Supplementary Fig.S6** [↗](#)). However, we observed a significant impairment in the entry of phage DNA into heat-treated bacteria (**Fig.4b** [↗](#)), suggesting a hindrance in the DNA injection process.

To explore this further, we employed liquid chromatography-mass spectrometry (LC-MS) for total protein analysis and identified that the down-regulated proteins were predominantly associated with the bacterial envelope (**Fig.4c, 4d** [↗](#)). In addition to a primary receptor, phages typically require a secondary receptor, often an outer membrane component (Sueki et al., 2020 [↗](#); Sun et al., 2006 [↗](#)), to establish adsorption and initiate DNA injection. Based on these findings, we hypothesized that heat treatment may induce the loss or modification of certain envelope components, thereby inhibiting phage DNA injection and contributing to heat-induced phage tolerance.

The bacterial envelope serves as a barrier against environmental stresses (Dehinwal et al., 2024 [↗](#); Mitchell & Silhavy, 2019 [↗](#); Silhavy et al., 2010 [↗](#)). We noticed that heat stress influenced not only the components mediating phage infection but also broader bacterial envelope dynamics, suggesting that overall envelope stress responses could be altered (Eberlein et al., 2018 [↗](#)). To investigate this, we examined the response of heat-treated bacteria to various envelope stresses, including unfavorable conditions such as pH changes and exposure to toxic compounds (metals, bile salts, and polymyxins), and found substantial changes in susceptibility (**Fig.4e** [↗](#)). Interestingly, heat-treated bacteria exhibited increased survival against polymyxins, similar to phage tolerance (**Figure 4e** [↗](#)). Notably, heat-treated bacteria showed significantly reduced lipopolysaccharide (LPS) production, known to mediate polymyxin adsorption (Yin et al., 2020 [↗](#)) (**Fig.4f** [↗](#)). Further experiments using the LPS inhibitor CHIR-090 confirmed that bacterial survival under polymyxin treatment increased with LPS reduction (**Fig.4g** [↗](#)). Given that polymyxin is a last-resort antibiotic for treating carbapenem-resistant *K. pneumoniae*, this cross-tolerance with phages might present a significant challenge to already scarce treatment options.

Phage Resistance in Heat-Evolved *pspA* Mutant

Heat treatment enhances bacterial tolerance to lytic phages, prompting us to investigate whether mutants surviving heat treatment could also withstand subsequent phage infections. To address this, we conducted a serial evolutionary experiment under selective heat stress (**Fig. 5a** [↗](#)). In line with our hypothesis, we identified two distinct mutants, BC2G11C1 and BC2G11C2, which exhibited significantly enhanced survival following heat treatment (**Fig. 5b** [↗](#)). However, only BC2G11C1 showed significantly reduced susceptibility to phage infection (**Fig. 5c** [↗](#)). Moreover, this strain maintained growth under phage treatment, indicating that the mutation conferred phage resistance (**Fig. 5d, 5e** [↗](#)). Thus, the BC2G11C1 mutant acquired both phage resistance and heat tolerance.

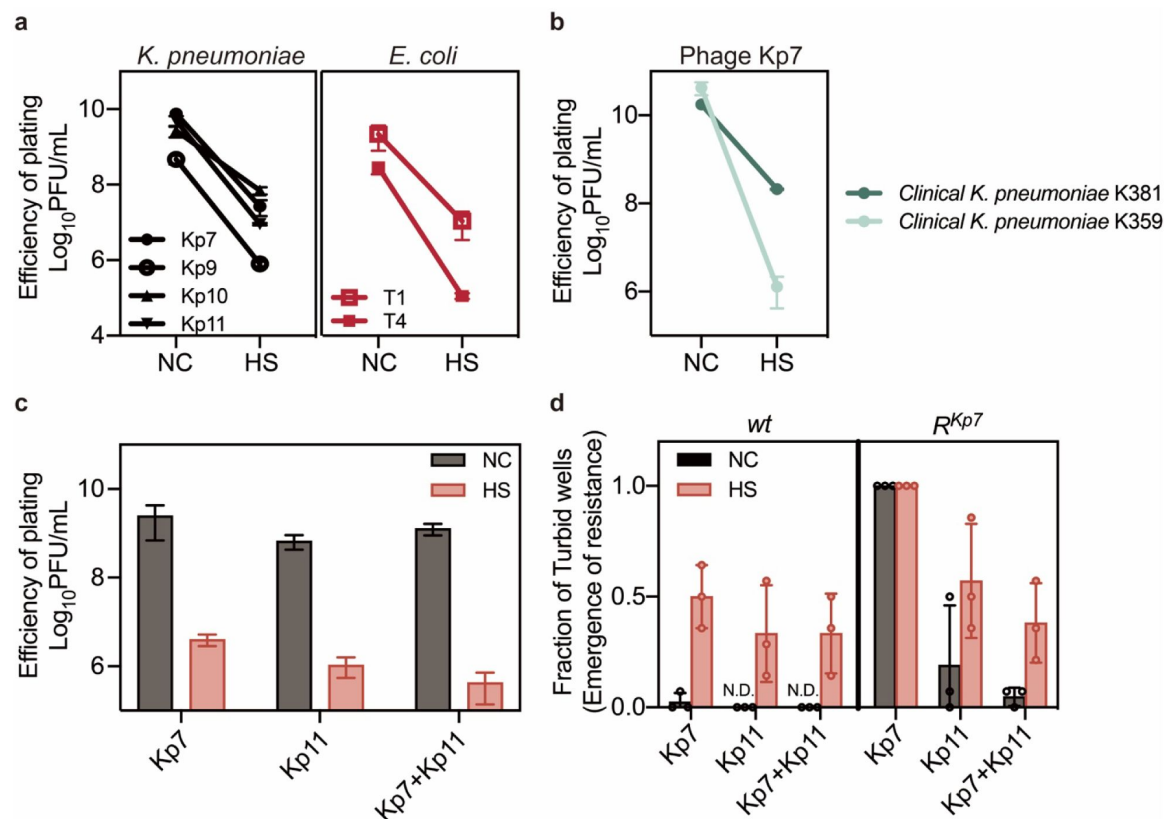


Fig.3

Heat-Induced Phage Tolerance Challenges Phage Therapy.

a, b. Small drop plaque assays were conducted using 10 - fold serial dilutions of phages Kp7, Kp9, Kp10 or Kp11 against *K. pneumoniae* ATCC 43816, as well as phage Kp7 against clinical *K. pneumoniae* strains K357 and K381, and phages T1 or T4 against *E. coli* KLY, both with and without heat treatment. Data are presented as mean $\pm$ s.d. of three independent replicates. **c.** Efficiency of plating assays were performed for individual phages or phage combination on *K. pneumoniae* ATCC 43816, with and without heat treatment. Data are presented as mean $\pm$ s.d. of three independent replicates. **d.** Bacteria ($\sim 10^3$ CFU) with and without heat treatment were cultured with either a single phage or phage combination at an MOI of 100 for 24 hours. The fraction of turbid wells indicates resistance emergence. Data are presented as mean $\pm$ s.d. of three independent replicates and 14 wells for each replicate.

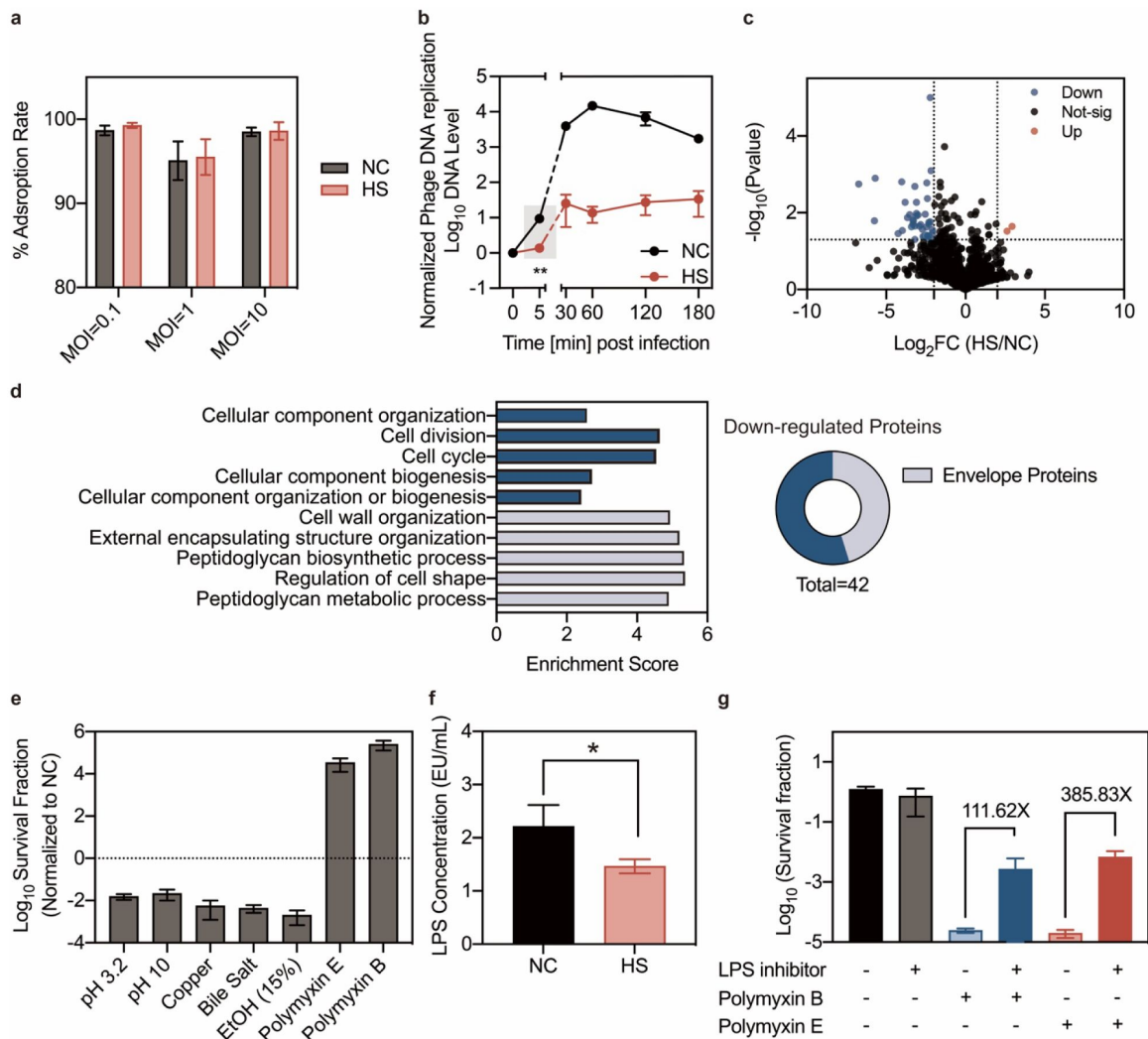


Fig.4

Reduced phage DNA entry and altered envelope stress responses in heat treated bacteria.

a. Adsorption assays of phage Kp11 to *K. pneumoniae* ATCC43816 were performed with and without heat treatment at various MOIs of 0.1, 1 and 10 for 3 min. Data are presented as mean $\pm$ s.d. of three independent replicates. **b.** The phage DNA levels in bacteria were measured at different time points. Phage Kp11 (MOI $\sim$ 100) were added at time 0. Data are presented as mean $\pm$ s.d. of three biological replicates. Statistical significance was determined using Significance levels were set at $**P < 0.01$ ($P = 0.0046$). **c.** Comparison of the protein profiling results for *K. pneumoniae* ATCC43816 with and without heat treatment. The color of each dot represents a P value calculated based on the log₂ ratio of protein abundance in the two populations (log₂(HS/NC)) (see Methods for the statistical test used). The best P values from all technical replicates are shown. The intensity is the normalized mean mass spectrometry peak intensity of each protein associated with the best P value among the three replicates. **d.** Gene ontology (GO) analysis of the downregulated proteins in biological process. **e.** Normalized survival fraction of heat-treated *K. pneumoniae* ATCC43816 under various stresses, including pH changes (pH 3.2 and pH 10 for 1 hour), copper (5mM CuSO₄ for 1 hour), ethanol (15% ethanol for 1 hour), bile salt (10mg/mL for 3 hours) and polymyxin (Polymyxin B, E 25 μ g/mL for 1 hour). Data are presented as mean $\pm$ s.d. of three independent replicates. **f.** The production of LPS in bacteria was quantified using the Chromogenic LAL Endotoxin Assay. Bars show mean $\pm$ s.d. of three biological replicates. Statistical significance was determined using Significance levels were set at $*P < 0.05$ ($P = 0.0377$). **g.** Survival fraction of *K. pneumoniae* treated with polymyxin for 1 hour, with or without the LPS inhibitor CHIR-090. Data are presented as mean $\pm$ s.d. of three independent replicates.

Whole-genome sequencing of the evolved strains revealed a truncation in the *pspA* gene in the phage-resistant mutant, which was responsible for the reduced sensitivity to both heat and phage treatment. In contrast, the other strain harbored a mutation in the *rpoH* gene, which conferred heat tolerance but did not affect phage resistance (**Fig. 5f**). The absence of similar phenotypes in the *pspA* knockout strain (**Supplementary Fig.S7**) suggested that this was a gain-of-function mutation in the *pspA* gene. This was further supported by the failure to restore the phenotypes upon reintroduction of the wild type *pspA* gene into the mutant strain; however, overexpression of the mutated *pspA* gene in the ancestral strain exhibited a similar phenotype (**Supplementary Fig.S7**).

Phage shock proteins, encoded by the *psp* operon, are reported to be induced by both heat stress and phage infection (Flores-Kim & Darwin, 2016; Model et al., 1997). However, the underlying mechanism has remained elusive. Our findings led us to hypothesize that the *psp* operon might function through envelope biogenesis, thereby linking responses to heat and phage treatment.

To test our hypothesis, we first investigated whether the phage life cycle in the *pspA** mutant was disrupted at an early stage, building on previous results regarding heat-induced phage tolerance. Adsorption assays revealed a significant impairment in phage adsorption to *pspA**, suggesting alterations in receptors availability (**Fig.6a**). Further analysis corroborated this observation, demonstrating a reduction in capsular production (**Fig.6b**) and a complete loss of capsular structure (**Fig.6c**).

Given the critical role of *K. pneumoniae* capsules in evading host immunity during septicemia (Huang et al., 2022), we assessed the immune evasion capacity of the *pspA** mutant. The results showed that *pspA** was rapidly cleared from mouse blood, demonstrating a significant reduction in immune evasion capacity (**Fig.6d** and **Supplementary Fig.S8**). A mild reduction in immune evasion was also observed in heat-treated bacteria (**Fig.6d**). Notably, the *pspA** mutant exhibited increased survival to polymyxin treatment (**Fig.6e**) without a corresponding increase in the minimum inhibitory concentration (MIC) (**Fig.6f**), suggesting a tolerant phenotype mediated by a reduction in LPS. Mass spectrometry analysis indicated that many down-regulated proteins in *pspA** were envelope-associated (**Fig.6g**), implying an important role for the *pspA* gene in maintaining envelope integrity.

Discussion

Previous reports have demonstrated that antibiotic persistence confers a distinct advantage to lysogenic phages but does not protect bacteria from lytic phages (Pearl et al., 2008). Here, we identified bacterial persistence against lytic phages, where bacteria survived extended phage treatment without developing resistant mutations. This introduces a novel strategy for bacterial defense against phages. Notably, our research revealed that heat stress enhances phage tolerance, subsequently promoting the emergence of resistance, aligned with observations in antibiotic research (Levin-Reisman et al., 2019; Levin-Reisman et al., 2017). This parallel emphasizes the necessity of cautious and strategic design of phage therapy. In line with recent work by Fernández-García et al. (Fernández-García et al., 2024), which showed that lytic phages can induce antibiotic persistence, our findings further highlight the existence of phage persistence, adding depth to bacterial survival strategies and reinforcing the complexity of phage therapy.

Heat, as a universal stress, induces protein misfolding and aggregation, despite the activation of heat shock proteins that mitigate these effects (Feder & Hofmann, 1999; Mogk et al., 2019). The accumulation of misfolded envelope proteins can become toxic, leading to cell death (Balchin et al., 2016; Konovalova et al., 2016; Takai et al., 1998). Correspondingly, the relatively low efficiency of translocation and assembly of envelope proteins (Bakkes et al., 2010; Dalbey & Kuhn, 2012; Vigh et al., 2007) renders them more vulnerable under heat treatment (Schiffman

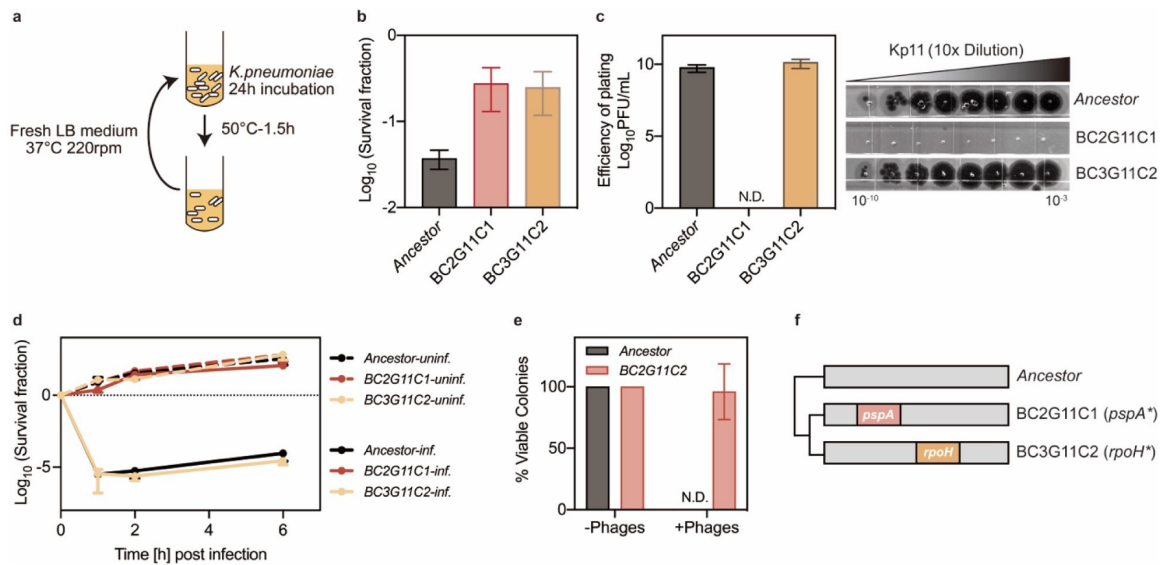


Fig.5

Mutation in *pspA* Confers Dual insensitivity to Thermal and Phage Stress.

a. Experimental design for cyclic exposure to heat treatment, where cultures of ATCC 43816 were subjected to a 50°C bath for 90 minutes. Following heat treatment, cultures were resuspended in fresh medium and incubated for 24 hours. **b.** Survival fraction of ancestral and evolved strains at 50°C for 90 minutes. Data are presented as mean $\pm$ s.d. of three independent replicates. BC: Batch Culture, G: Evolution Cycle Number, C: Colony. BC2G11C1 refers to the first colony from batch culture 2 after 11 rounds of heat treatment. **c.** Efficiency of plating assay for phage Kp11 to ancestral and evolved strains. Data are presented as mean $\pm$ s.d. of three independent replicates. **d.** Survival fraction of ancestral and evolved strains from phage Kp11 infection (MOI ~ 100). Curves show the mean $\pm$ s.d. of three independent replicates. **e.** Viable colonies of ancestral and evolved strains grown on both phage plates (10⁸ PFU) and phage-free plates. Data are presented the mean $\pm$ s.d. of three independent replicates. **f.** Nonsynonymous mutations identified in the strains isolated from the evolved culture.

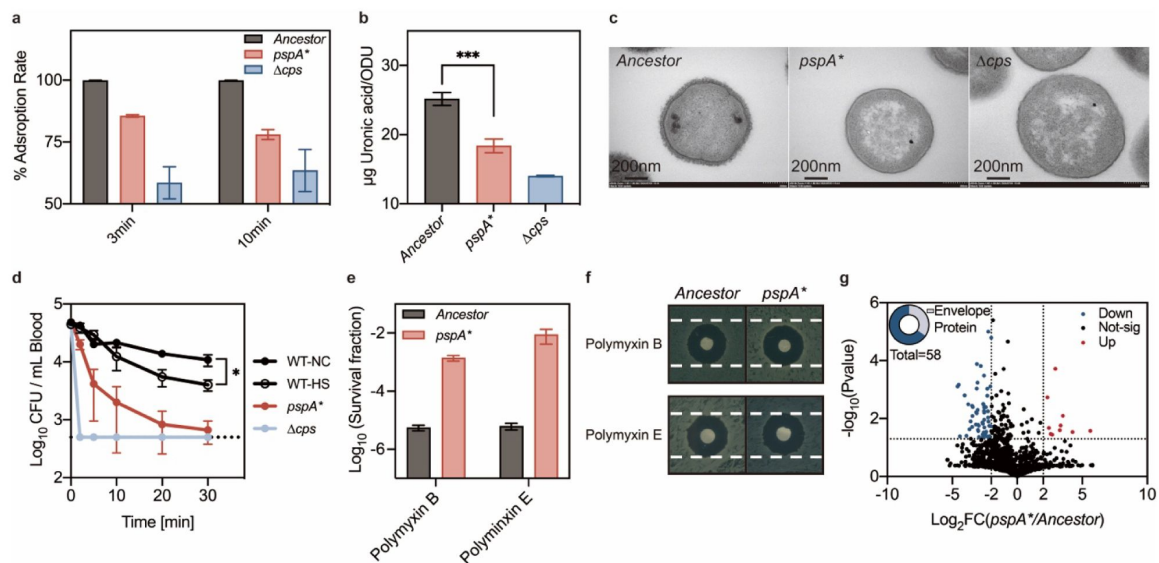


Fig.6

***pspA* Mutation Impaired Phage Adsorption and altered Envelope Stress Responses.**

a. Adsorption assay of phage Kp11 to the ancestral strain *K. pneumoniae* ATCC 43816, *pspA** and the non-capsular strain *Dcps* at an MOI of 0.1. **b.** Quantification of capsular production of the ancestral strain, *pspA** and *Dcps*. The capsular polysaccharide (CPS) was quantified using the uronic acid method, presented as the amount of uronic acid per OD600. Data are presented the mean $\pm$ s.d. of three independent replicates; statistical significance was assessed using an unpaired two-tailed t-test, with statistical significance levels set at *** $P < 0.005$ ($P = 0.001$). **c.** Detection of bacterial capsular structure using transmission electronic microscopy (TEM). **d.** Virulence traits were assessed by intravenously (i.v.) infection of ICR mice with $\sim 10^6$ of each bacterial strain (ancestral with and without heat treatment, *pspA** and *Dcps*) to determine bacteremia levels. Curves show the mean of three independent replicates, error bars are s.d. of the mean. Statistical significance was assessed using an unpaired two-tailed t-test, with significance levels set at * $P < 0.05$ ($P = 0.011$). **e.** Survival fraction of bacteria (ancestral and evolved strain *pspA**) after polymyxin treatment (polymyxin B and E, 25 μ g/mL for 1 hour). Data are presented the mean $\pm$ s.d. of three independent replicates. **f.** Disk fusion assay (polymyxin B and E, 10 μ g) of the ancestral and evolved strain *pspA**. **g.** Comparison for the protein profiling results for the *ancestors* and *pspA**. The color of each dot represents a P value calculated based on the $\log_2$ ratio of protein abundance in the two populations ($\log_2(pspA^*/Ancestor)$).

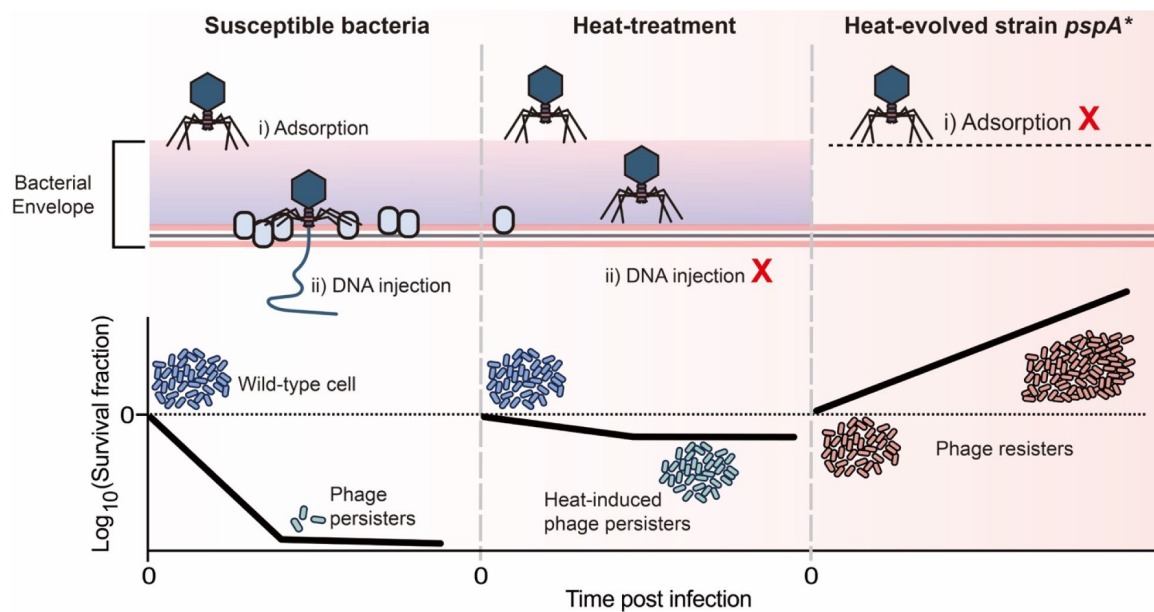


Fig.7

Schematical Model of this work.

A small proportion of phage-sensitive bacteria can survive during lytic phage infections, exhibiting persistence against phage. Heat treatment systematically reduces envelope-associated proteins in bacteria, inhibiting the entry of phage DNA and thereby inducing phage tolerance. Mutation in the gene *pspA* resulted in the loss of capsule, conferring receptor-deficient phage resistance.

et al., 2017 [↗](#)), resulting in a decrease in abundance. This reduction in envelope protein might be an adaptive response to elevated temperatures, as evidenced by the observations following heat treatment. Consequently, this would lead to a decrease in phage receptors (such as capsule and outer membrane protein FhuA) and the receptor for polymyxin (LPS), thereby conferring tolerance. Recent studies have shown that heat shock potentiates aminoglycosides against Gram-negative bacteria by enhancing antibiotic uptake and promoting protein aggregation (Lv et al., 2023 [↗](#)), indicating that heat stress can contribute to bacterial sensitivity to antibiotics, which could also extend to phages in our study.

This phenomenon may also occur in the heat-tolerant mutant *pspA*^{*}, where the reduction in envelope components resulted in insusceptibility to phages and polymyxins, highlighting a potential regulatory balance governed by PspA. PspA, a key regulator in the phage shock protein system, functions as part of the envelope stress response system in bacteria, preventing membrane depolarization and ensuring the envelope stability (Flores-Kim & Darwin, 2016 [↗](#); Horstman & Darwin, 2012 [↗](#); Kleerebezem & Tommassen, 1993 [↗](#); Yamaguchi et al., 2010 [↗](#)). Recent studies have shown that PspA is analogous to ESCRT-III, a component of the ESCRT machinery in eukaryotic cells, which plays a crucial role in membrane remodeling (Junglas et al., 2021 [↗](#); Liu et al., 2021 [↗](#)).

The bacterial envelope acts both as a defense barrier against external stresses and as a conduit for invaders. While it reduces the invasion of harmful substances and maintains intra-extracellular balance (Arts et al., 2015 [↗](#); Raivio, 2014 [↗](#); Wood, 2015 [↗](#)), it also provides contacts for invaders; for example, capsules serve as receptors for phages (Latka et al., 2021 [↗](#)), and LPS interacts with polymyxins (Manioglu et al., 2022 [↗](#)). Thus, heat-induced alteration to the envelope could be a double-edged sword. Clinically, heat stress would not only impact phage susceptibility but also alters the interaction between *K. pneumoniae* and the host immune system. This dual impact underscores the complexity of bacterial adaptive responses and suggests that environmental factors can significantly influence the outcome of phage therapy. Furthermore, increased sensitivity to acidic conditions and bile salts induced by heat stress challenges bacterial survival, particularly relevant to *K. pneumoniae* colonization sites in the human body (Gasink et al., 2009 [↗](#); Martin et al., 2016 [↗](#)).

Lastly, the envelope stress responses induced by heat in bacteria lead us to consider the role of fever. We found that heat-induced phage tolerance is both time- and temperature-dependent; even moderately elevated temperatures, such as 42°C, can induce phage tolerance with prolonged treatment, indicating alterations in the bacterial envelope. This suggests that elevated body temperature, akin to fever, may affect bacterial survival and colonization by modulating their responses to the host immune system and antimicrobial agents. Since elevated temperatures also alter life-history traits in phage communities (Greenrod et al., 2024 [↗](#)), they may similarly impact bacterial responses to heat stress, phage, and antibiotic treatments, underscoring the need for further research into the role of fever in bacterial growth and survival during therapy.

Materials and Methods

Bacteria, Phages and Plasmids

K. pneumoniae and *E. coli* strains used in this work are listed in **Table S1** [↗](#). *E. coli* strain *DH5D* was used for DNA cloning (Seidman & Struhl, 2001 [↗](#)). The phages used in this work are also listed in **Table S1** [↗](#). Plasmid constructions were performed in *E. coli* strain *DH5⁺* using standard methodologies and are listed in **Table S2** [↗](#).

Phage purification & concentration

Overnight cultured bacteria were inoculated into 200 mL of fresh LB medium and grown to an OD_{600} of ~ 0.3 . Then, 100 μ L of phage stock was added to the cultures for a few hours, after which the cultures were centrifuged at 4°C, at 4000 rpm when cultures turn to transparency. The supernatants were filtered using a 0.22 μ m filter to wipe off bacteria, and the phage cultures were concentrated using ultracentrifuge tubes (100 KD). Filtered SM buffer was added to the concentrated phage cultures until a stable volume was achieved, and the final phage stocks were stored at 4°C (Bonilla et al., 2016 [↗](#)).

Killing dynamic of phage infection and phage persisters definition

Killing dynamic of phage infection to *K. pneumoniae* were performed by treating cells ($\sim 10^6$ CFU/mL) with lytic phage Kp11 (MOI ~ 100) for 120 minutes while shaking at 220 rpm in a 15 mL tube. Samples were taken at indicated time points, washed three times with PBS buffer and CFUs were evaluated by plating on LB agar plates before and after phage infection. To determine the persistence fraction and obtain phage persisters, a single colony from post-phage treatment was inoculated into fresh LB broth and grown at 37°C with shaking at 220 rpm for 6 hours. Then each culture was collected and dropped on solid LB agar plates supplemented with and without phages. Phage persisters were identified as those solely capable of growth on phage-free plates, with the fraction of persisters estimated by subtracting the number of resistant mutants from the total CFUs. After identification, descendants of persisters were diluted to $\sim 10^6$ CFU/mL and infected with phage Kp11 for 40 minutes, with CFUs evaluated by plating before and after phage treatment over several cycles.

Plaque assays

Phage titers and efficiency of plating were determined using the small drop plaque assay method (Goldfarb et al., 2015 [↗](#)). 200 μ L of bacterial culture was mixed with 5 mL of melted 0.7 % soft LB agar medium and overlaid on 1.5 % solid LB plates to form bacterial lawn. After allowing the bacterial lawn to dry for 15 minutes, 5 μ L of 10-fold serial dilutions of phages in SM buffer were dropped to the lawn. Once the drops had dried, the plates were inverted and incubated at 37°C overnight. Plaque forming units (PFUs) were determined by counting the derived plaques after incubation, with lysate titers calculated as PFUs per mL. If no individual plaques could be observed, a faint lysis zone across the drop area was considered equivalent to 10 plaques. The efficiency of plating (EOP) was measured by comparing results from control and treated bacteria.

ScanLag analysis

Bacteria were diluted serially and plated on solid LB agar medium. The plates were placed in a ScanLag (Levin-Reisman et al., 2010 [↗](#)) setup at 37°C, which consists of an array of office scanners that automatically capture images of the plates every 20 minutes, thus monitoring the appearance of colonies. An automated image analysis application extracts the distribution of appearance time of the colonies.

Growth dynamic of phage infection culture

In this work, bacterial cultures with and without heat treated were diluted to 10^5 - 10^6 CFU/mL in LB medium and infected with high concentration of phages at an MOI of 100. Optical density measurements at a wavelength of 600nm were taken every 10 minutes using a TECAN Infinite 200 plate reader in a 96-well plate.

Fluctuation test for measuring mutation rate (modified from Delbrück-Luria Fluctuation test)

To measure the mutation rate of heat-treated bacteria to phage, several replicas of culture (~1000 bacteria per well) were grown overnight in a 96-well plate to 10^9 CFU/mL in 200 μ L and treated with and without heat stress. Then each culture was plated onto solid agar plates supplemented with 10^9 PFUs of phages and incubated at 37°C. The fraction of empty plates was used to calculate the mutation rate from five biological repeats of 14 replicas (Luria & Delbrück, 1943 [\[1\]](#)).

Resistance emergence fraction experiments

Bacteria culture was seeded in 96-well plate to achieve an approximate total bacterial load of 1000 CFU per well, minimizing the impact of spontaneous resisters and ensuring that any observed resistance was due to division during phage infection. Bacteria in each well were incubated with phages at a various MOI for 24 hours. Given the exceptionally low total bacterial load utilized in this experiment, the potential for phage-resistant bacteria due to stochastic mutations would be minimized. The high MOI guaranteed that each bacterium was likely to be infected by at least one phage particle. Due to the rapid infection and lysis in this system, the probability of observing turbid wells after 24 hours was minimal. The fraction of turbid wells was used to calculate the phage resistance emergence from three biological repeats of 14 replicates.

Time-lapse microscopy and image acquisition

To observe bacterial dynamics under phage treatment, time-lapse microscopy was performed using an FV3000 with a temperature-controlled stage. *K. pneumoniae* cultures with and without heat treatment were exposed to lytic phage Kp11 at an MOI of 100. For time-lapse imaging, bacterial cultures were transferred to a 35 mm glass-bottom dish (MatTek Corporation) and incubated at 37°C in a humidified atmosphere. The microscope was set to capture images at regular intervals (e.g., every 2 minutes for un-treated bacteria and 10 minutes for heat-treated bacteria). A 100x oil immersion objective was used to obtain high-resolution images. Images were captured using Ti2-E (Nikon, Japan) with a live-cell system FV3000 RS (Olympus, Japan). The resulting image stacks were analyzed using ImageJ.

Phage adsorption assay

The indicated bacterial cultures were diluted to fresh LB medium and infected with phages at different MOIs (0.1, 1, 10). 500 μ L of samples were collected at 3- and 10 minutes post-infection and centrifuged for 1 min at 12000 rpm. 500 μ L of LB broth with 50 μ L of chloroform were added to the supernatant, which was then placed on ice for 10 min to remove bacterial body. Phage adsorption to the indicated bacteria was measured by titrating the free phages present in supernatant (Kim et al., 2019 [\[2\]](#)).

Transmission Electron Microscopy

The samples were fixed in a mixture of 2% paraformaldehyde and 2.5% glutaraldehyde, followed by four washes with 0.1 M phosphate buffer (PB). Post-fixation was performed using 1% osmium tetroxide and 1.5% tetrapotassium hexacyanoferrate trihydrate for 1 hour at 23°C. Samples were then dehydrated through graded ethanol solutions (50%, 70%, 80%, 90%, 100%, 100%, 100%) for 15 minutes each. Next, they were infiltrated twice with 1,2-epoxypropane for 15 minutes each, followed by gradient infiltration with a mixture of 1,2-epoxypropane and Epon 812 resin for 8 hours (SPI, America). The samples were then embedded in pure Epon 812 twice and polymerized in an oven at 60°C. Blocks of polymerized resin were sectioned using a Leica EM UC7 ultramicrotome (Wetzlar, Germany), yielding ultra-thin sections (70 nm) mounted on coated

copper grids. Sections were stained on-grid with 2% uranyl acetate for 25 minutes and lead citrate for 5 minutes. Imaging was conducted using an H-7650B transmission electron microscope (Hitachi, Tokyo, Japan).

Phage DNA injection and replication kinetics monitoring

Indicated bacteria were infected with phages at an MOI of 10. At indicated time points, cells were harvested, and total DNA (including both bacterial and phage DNA) was extracted. qPCR was performed using primers specific to the phage genome (shown in [Table S2](#)), allowing quantification of phage DNA levels by comparing qPCR results.

Mass spectrometry

10 mL of bacteria for each sample were used for total protein extraction by using Bacterial Protein Extraction Kit (C600596, Sangon) and subjected to liquid chromatography-mass spectrometry (LC-MS). Briefly, proteins were disulfide reduced with 25 mM dithiothreitol (DTT) and alkylated with 55 mM iodoacetamide. In-gel digestion was performed using sequencing-grade modified trypsin in 50 mM ammonium bicarbonate at 37 °C overnight. The peptides were extracted twice with 1% trifluoroacetic acid in 50% acetonitrile for 30 min. The peptide extracts were then centrifuged using a SpeedVac. For LC-MS/MS analysis, peptides were separated by a 60-minute gradient elution at a flow rate 0.300 μ L/min with a Thermo-Dionex Ultimate 3000 HPLC system, directly interfaced with the Thermo Orbitrap Fusion mass spectrometer. The analytical column was a homemade fused silica capillary column (75 μ m ID, 150 mm length; Upchurch, Oak Harbor, WA) packed with C-18 resin (300 A, 5 μ m; Varian, Lexington, MA). Mobile phase A consisted of 0.1% formic acid, and mobile phase B consisted of 100% acetonitrile and 0.1% formic acid. The Orbitrap Fusion mass spectrometer was operated in the data-dependent acquisition mode using Xcalibur 3.0 software and there is a single full-scan mass spectrum in the Orbitrap (350-1550 m/z, 120,000 resolution) followed by 3 seconds data-dependent MS/MS scans in an Ion Routing Multipole at 30% normalized collision energy (HCD). The MS/MS spectra from each LC-MS/MS run were searched against the selected database using the Proteome Discovery searching algorithm (version 1.4).

Evolutionary protocol of cyclic exposure to heat stress

The cyclic evolution protocol ([Figure 5](#)) consists of two steps: killing by heat stress at 50 °C for 90 minutes and 24 hours growth. Three replicate populations evolved, each initiated from a single colony isolated from the ancestral glycerol stock in 2 mL of fresh LB medium, which was shaken at 220 rpm at 37 °C for 24 hours. The cultures were transferred to an elevated temperature of 50 °C for 90 minutes. Then the cultures were resuspended in 2 mL of fresh LB medium and grown for 24 hours at 37 °C with shaking. Half of the regrown cultures were used for analysis. After each cycle, samples from each culture were diluted serially and plated on solid LB agar medium to determine the survival.

Mouse Infection

Bacteremia modeling in mice followed the approach described by Huang et al ([Huang et al., 2022](#)). Female ICR mice (6-8 weeks old) were intravenously (i.v.) injected with *K. pneumoniae*. Bacteria were collected from pre-counted frozen stocks and resuspended in Ringer's solution to the desired concentration for injection (100 μ L). To calculate actual infectious doses, bacterial CFUs were plated one day prior to infection. Bacterial levels in the blood were measured by orbital hemorrhage at set time points post-infection, with the initial bacterial load (0 min) determined by dividing inoculum CFUs by blood volume (0.07 mL/g of body weight). Bacterial counts in organs were measured similarly, using tissue homogenates and estimating residual blood volume at 0.1 mL per gram of organ. The total bacterial burden was calculated by summing the CFU counts in the blood and organs, and survival rate was determined by dividing the total surviving bacteria by the initial inoculum CFUs.

Genomic extraction and whole-genome sequencing

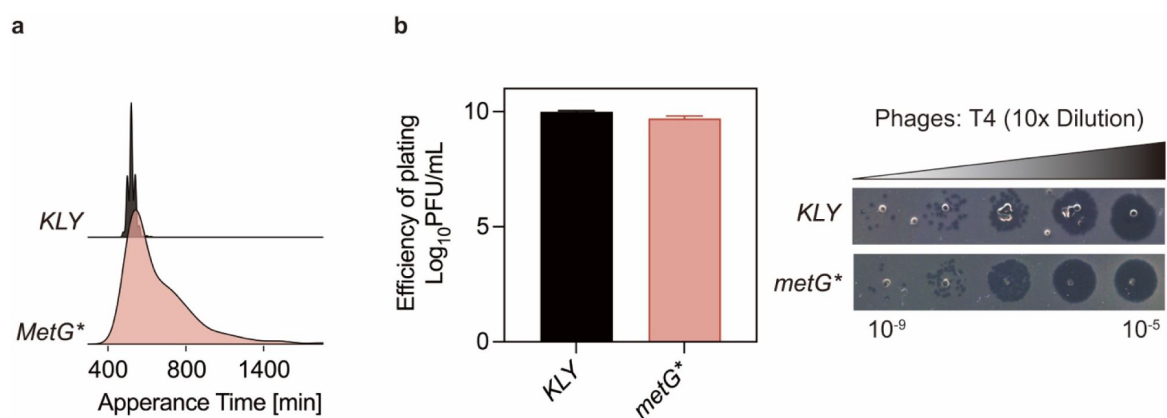
Genomic DNA was extracted using the Tianamp Bacteria DNA Kit (Tiangen). Sequencing was performed by Sangon using the Illumina-HiSeq with 150 bp paired-end reads, at an average coverage of 100-200. Identification of single nucleotide polymorphisms (SNPs) was performed by aligning the raw Illumina reads to the genome of the reference strain ATCC 43816 using bwa version 0.7.18.

Statistical Analysis

All experiments were performed with at least three independent biological replicates unless otherwise specified. Bar graphs display the mean $\pm$ standard deviation (s.d.). Unpaired, two-tailed t-tests were used to calculate P values for comparisons of survival rates, EOP, and LPS production, with statistical significance set at $P < 0.05$. For the protein profiling experiments, differential protein abundance between the heat-treated and control groups were analyzed using Student's t-tests for each protein. Multiple comparisons were adjusted using the Benjamini-Hochberg false discovery rate (FDR) correction to control for Type I errors. Proteins with a corrected P-value < 0.05 were considered significantly differentially expressed.

Supplementary Materials

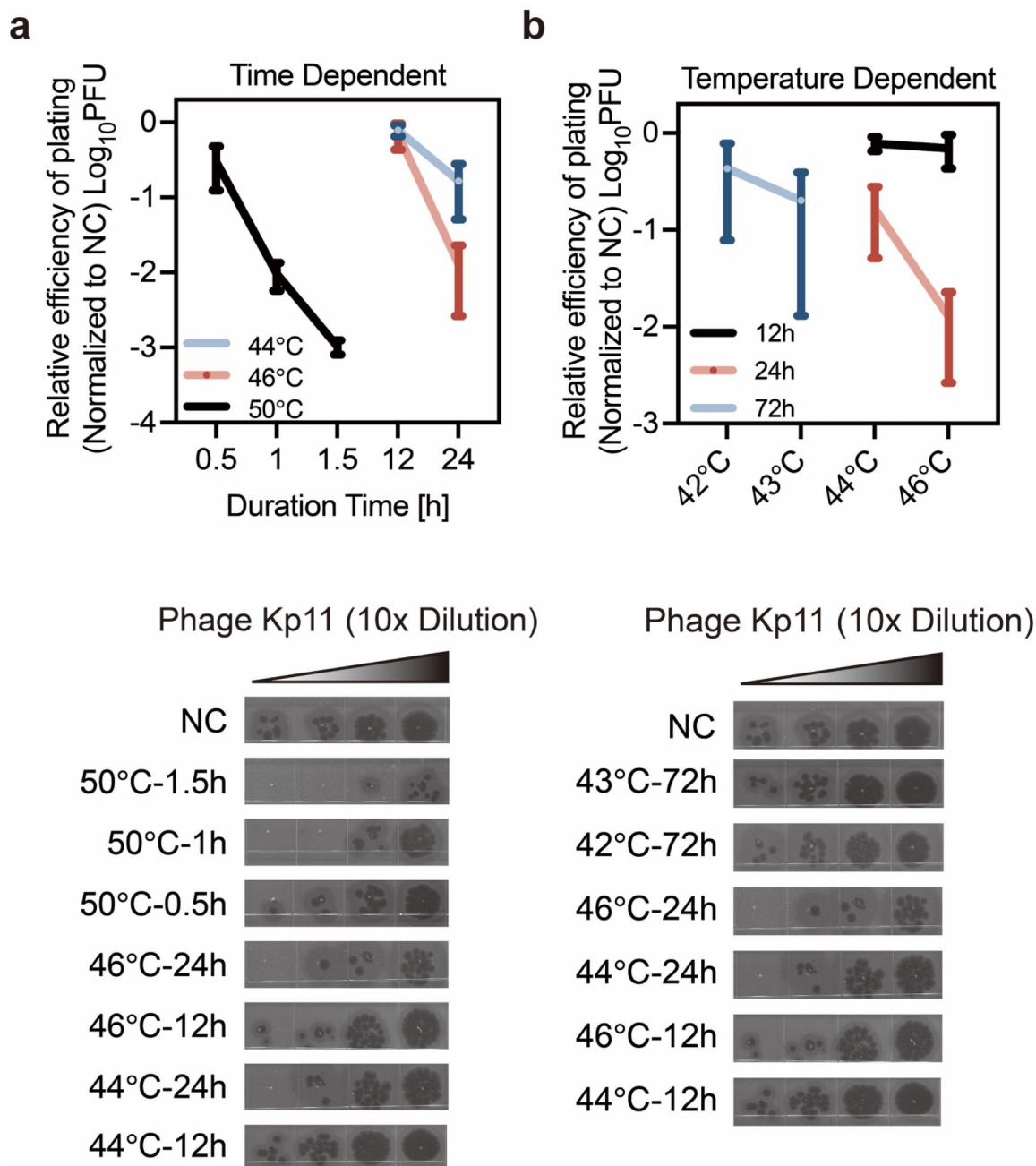
Supplementary Figures



Supplementary Fig.S2.

Susceptibility to phage T4 is similar for *KLY* and longer lag-time strain *metG.**

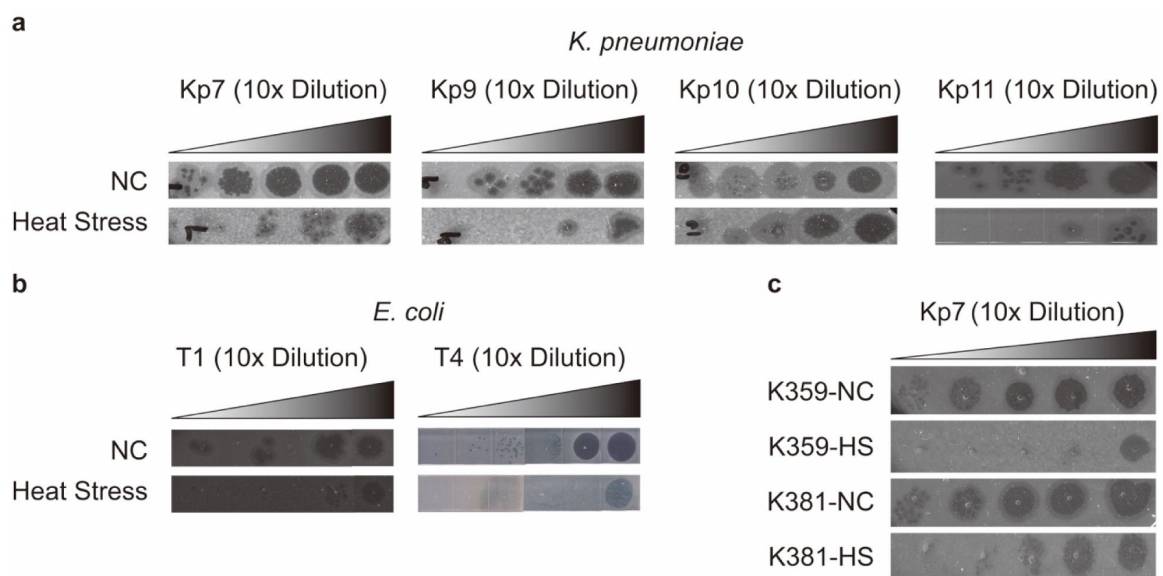
a. The distribution of appearance times of colonies for *E. coli* strain *KLY* and *metG** was measured using ScanLag (refer to Materials and Methods). The y-axis represents the normalized proportion of colony-forming units (CFUs) detected at each time points. **b.** The relative minimum phage titer of *KLY* and *metG** for T4 test and small drop assay phenotypes. Data are presented as the mean $\pm$ s.d. from three independent biological replicates. This data corresponds to the results discussed in [Fig.1](#) of the main text, showing increased lag time does not confer protection against phage infection.



Supplementary Fig.S3.

Heat induced phage tolerance was impacted by treatment time and temperature.

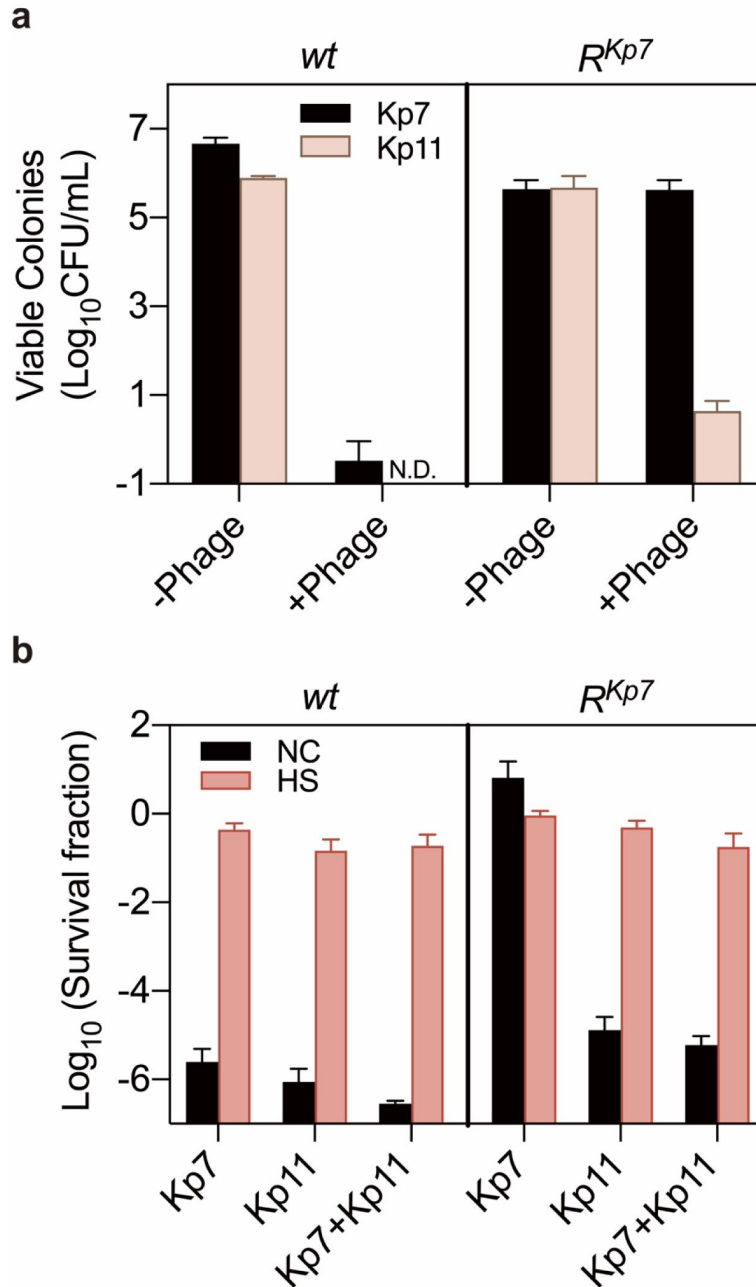
Phage tolerance in *K. pneumoniae* induced by heat stress as a function of treatment duration (**a**) and treatment temperature (**b**). Data are presented as mean $\pm$ s.d. from three independent biological replicates. This data corresponds to the results discussed in **Fig.2** of the main text, showing effects of heat-treated phage tolerance being both time- and temperature-dependent.



Supplementary Fig.S4.

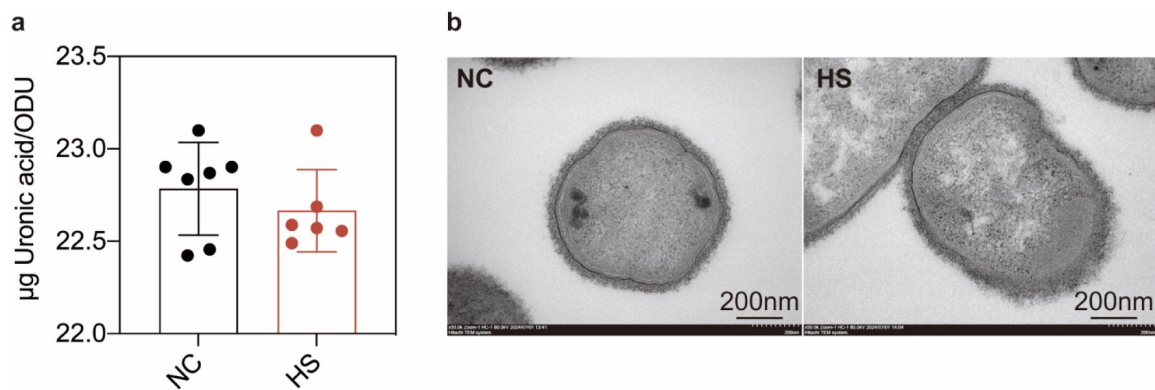
Phage tolerance induced by heat stress is a consistent response observed across various bacterial strains.

Morphological comparison of *K. pneumoniae* ATCC 43816 infected with phages Kp7, Kp9, Kp10, and Kp11 (**a**), *E. coli* KLY infected with phages T1 and T4 (**b**), and clinical *K. pneumoniae* strains K359 and K381 infected with phage Kp7 (**c**), before and after heat treatment. This data corresponds to the results discussed in **Fig.3** of the main text.



Supplementary Fig.S5.

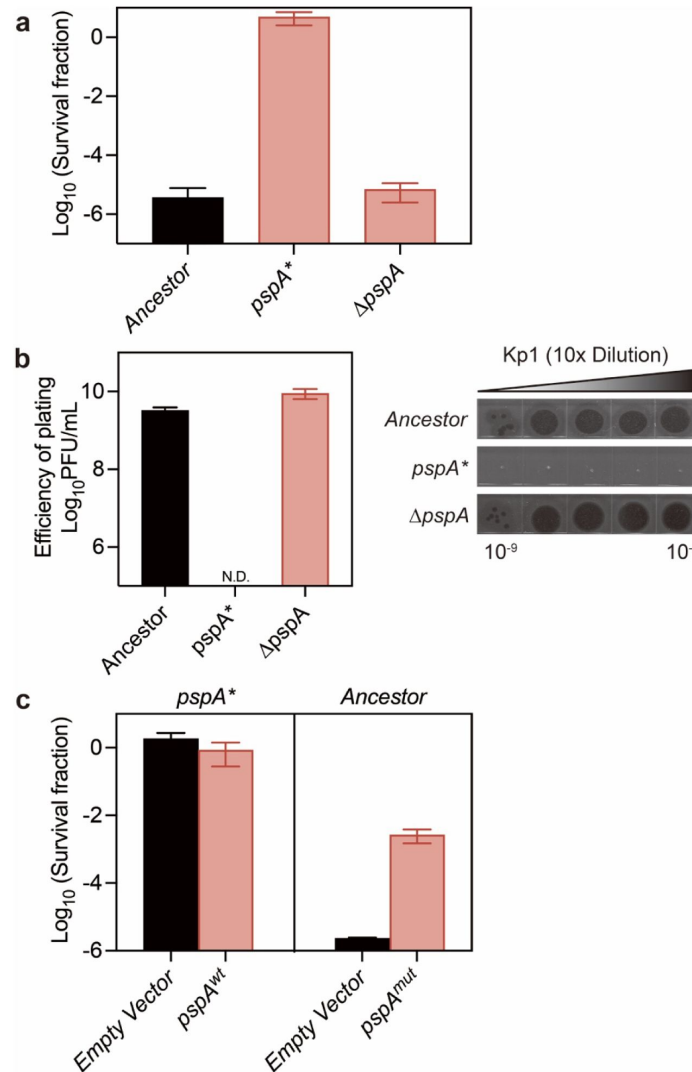
a. The Kp7-resistant strain (R^{Kp7}) remained sensitive to phage Kp11. Colonies of R^{Kp7} were detectable only on the Kp7 phage plates but not on the Kp11 phage plates, while the wild-type (*wt*) strain was sensitive to both phages, showing no viable colonies on either plate. Data are presented as the mean $\pm$ s.d. independent biological replicates. **b.** Killing assay of *wt* and R^{Kp7} strains with phage Kp7, Kp11, and their combination, before and after heat treatment. Heat stress increased bacterial survival under all conditions, including exposure to Kp7 alone, Kp11 alone, and the combination of Kp7 and Kp11. Data are presented as the mean $\pm$ s.d. from three independent biological replicates. This data corresponds to the results discussed in Fig.3 of the main text.



Supplementary Fig.S6.

Production and structure of the bacterial capsule in heat-treated and untreated bacteria.

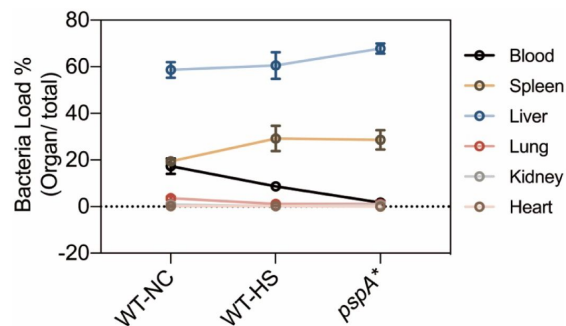
a. Quantification of capsular production of untreated bacteria and heat-treated bacteria. Capsular production was measured using the uronic acid method and expressed as the amount of uronic acid per OD600. Data are presented as the mean $\pm$ s.d. from six independent biological replicates. **b.** Detection of bacterial capsular structure by transmission electronic microscopy (TEM) before and after heat treatment. This data corresponds to the results discussed in [Fig.4](#) of the main text, showing the production and structure of the bacterial capsule remained unchanged.



Supplementary Fig.S7.

Impact of *pspA* mutations on bacterial survival and phage resistance.

a.b. The *pspA* knockout strain did not display the phenotype associated with the *pspA* mutation in survival (**a**) and efficiency of plating (**b**) when exposed to phage Kp11. **c.** Reintroduction of wild-type PspA into the mutant strain did not restore the original phenotype; However, overexpression of the mutant PspA enhanced survival during phage infection. Data are presented as the mean $\pm$ s.d. from three biological replicates. This data corresponds to the results discussed in **Fig.5** of the main text, showing the mutant in gene *pspA* was a gain-of-function mutation.



Supplementary Fig.S8.

Superior protection of un-treated ancestral strain ATCC 43816 against immune clearance of septic infection.

Female ICR mice were intravenously (*i.v.*) infected with approximately 10^6 CFU of the ancestral strain ATCC 43816, with and without heat treatment, and the *pspA** mutant. CFU counts in the blood and major organs were determined at 30 minutes post-infection. Data are presented as the mean $\pm$ s.d. from three independent biological replicates. This data corresponds to the results discussed in [Fig.6](#) of the main text, showing a significant reduction in immune evasion capacity both in *pspA** and heat-treated ancestral strain.

Supplementary Tables

Strain Name	Description	Reference
<i>K. pneumoniae</i>		
ATCC 43816		
<i>K381</i>	Isolated from Hospital	This Study
<i>K357</i>	Isolated from Hospital	This Study
<i>pspA*</i>	Evolution BC2G11C1	This Study
<i>rpoH*</i>	Evolution BC3G11C2	This Study
<i>Δcps</i>	ATCC 43816 <i>Δcps</i>	(Huang et al., 2022)
<i>ΔpspA</i>	ATCC 43816 <i>ΔpspA</i>	This Study
<i>E. coli</i>		
<i>KLY</i>	KL16-YFP Cam	(Fridman et al., 2014)
<i>metG*</i>	<i>KLY metG^T</i>	(Levin-Reisman et al., 2017)
Phages		
Kp11	<i>K. pneumoniae</i> phage (GenBank: ON148528.1)	Provided by Xiang Lab.(Huang et al., 2024)
Kp7	<i>K. pneumoniae</i> phage (GenBank: ON148527.1)	
Kp9	<i>K. pneumoniae</i> phage (GenBank: ON148529.1)	
Kp10	<i>K. pneumoniae</i> phage	
T1/T4	MG 1655 <i>E. coli</i> phage	Provided by Liang Lab.

Table S1

List of bacterial strains and phages used in this study.

Table S2

List of plasmids and primers used in this Study.

Primer name	Description	Reference
pCasKP-apr	Thermosensitive plasmid carrying Cas9 gene and lambda red system; Apr ^R	(Wang et al., 2022)Provided By Ji Lab.
pSGKP-spe	Expressing sgRNA and carrying <i>sacB</i> ; Spe ^R	
Primers		
ZF-pspA1	tagtGCAGTTTGCCGAACTGAAAGCGG	This Study For pspA deletion
ZF-pspA2	aaacCCGCTTTCAGTTCGGCAAACCTGC	
ZF-pspA3	TGTCGCCAGCTCATCGAGAAACAGCGTC	
ZF-pspA4	TTTCTCGATGAGCTGGCGACA	
ZF-pspA5	TGAAACCAGAACTATGAGGATTGAAA TTGTATGATTGCGCGGGCGCGGCGTT GCG	
ZF-pspA6	ATGAGCTCCACCGCGGTGGCGGGCCG CTCTAGCCGTACAGCTCGCTGTCCG GTAT	
ZFqP1-F	GACGATCCCTAGCTGGTCTG	This Study For qPCR
ZFqP1-R	GTGCAATATTCCTCCACTGCT	
ZFqP2-F	GACATCCCGTTTCGACTACCT	
ZFqP2-R	GCAGCTGTGAGACGTTAAGG	

Table S3

Mutation rate of *K. pneumoniae* to phage.

	replicates	Nt (x10 ⁷)	No mutation cultures	P ₀	Mutation rate
NC	1	23	14/14	1	1.23x10 ⁻⁹
	2	3.7	12/14	0.875	
	3	11	13/14	0.929	
	4	5.6	13/14	0.929	
	5	1.8	14/14	1	
HS	1	1.9	12/14	0.875	3.61x10 ⁻¹⁰
	2	2.3	14/14	1	
	3	2.8	11/14	0.786	
	4	1.8	12/14	0.875	
	5	1.1	14/14	1	

Data and materials availability

All data needed to evaluate the conclusions in the paper are present within the paper and/or the Supplementary Materials.

Acknowledgements

We thank Quan-Jiang Ji (ShanghaiTech University) for providing plasmids used in *K. pneumoniae* genome editing and Nathalia. Q. Balaban (Hebrew University of Jerusalem) for the *E. coli* strains *KLY* and *metG**. We are grateful to Guan-Xiang Liang, Yao-Yu Yang, and Gao-Pu Zhang (Tsinghua University) for the phages, and Jun-Hao Zhu (Institute of Microbiology, Chinese Academy of Sciences) for manuscript comments. We acknowledge Hai-Teng Deng and Xian-Bin Meng (Proteomics Facility, Technology Center for Protein Sciences, Tsinghua University) for assistance with protein MS analysis, and Jing-Jing Wang (Cell Biology Facility, Center for Biomedical Analysis, Tsinghua University) for technical support. We also thank the Cell Biology Facility (Center for Biomedical Analysis, Tsinghua University) for equipment support.

Additional information

Funding

National Key Research and Development Program of China (2022YFC2303202);

Tsinghua-Peking Joint Center for Life Sciences (20111770319) ;

Tsinghua University Dushi Program (20231080044).

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Supervision: J. -F. Liu

Writing-original draft: F. Zhang and J. -F.

Liu Writing-review & editing: J.-R. Zhang and Y. Xiang

Additional files

Supplementary Video 1. Untreated bacteria were lysed by phages before cell division could occur. Shown is the rapid lysis of untreated *K. pneumoniae* ATCC43816 infected with phage Kp11 at an MOI of 100. The bacteria are lysed before they can undergo cell division, demonstrating the efficiency of phage-induced

lysis at this infection level. Watch as bacteria are quickly disrupted following phage exposure. [🔗](#)

Supplementary Video 2. Heat-treated bacteria were able to undergo cell division despite exposure to the lytic phage. Shown is the infection of heat-treated *K. pneumoniae* ATCC43816 by phage Kp11 at an MOI of 100. Unlike the untreated bacteria, these bacterial cells can be observed undergoing division before lysis occurs. The delayed lysis highlights the increased tolerance of heat-treated bacteria to phage infection under similar conditions. This increased frequency for bacterial division facilitates the evolution of phage resistance, highlighting the impact of heat treatment on bacterial survival under phage infection. [🔗](#)

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Reviewer #1 (Public review):

Summary:

In this interesting and original paper, the authors examine the effect that heat stress can have on the ability of bacterial cells to evade infection by lytic bacteriophages. Briefly, the authors show that heat stress increases tolerance of *Klebsiella pneumoniae* to infection by the lytic phage Kp11. They also argue that this increased tolerance facilitates the evolution of genetically encoded resistance to the phage. In addition, they show that heat can reduce the efficacy of phage therapy. Moreover, they define a likely mechanistic reason for both tolerance and genetically encoded resistance. Both lead to a reorganization of the bacterial cell envelope, which reduces the likelihood that phage can successfully inject their DNA.

Strengths:

I found large parts of this paper well written and clearly presented. I also found many of the experiments simple yet compelling. For example, the experiments described in figure 3 clearly show that prior heat exposure can affect the efficacy of phage therapy. In addition the experiments shown in figure 4 and 6 clearly demonstrate the likely mechanistic cause of this effect. The conceptual figure 7 is clear and illustrates the main ideas well. I think this paper would be publishable even without its central claim, namely that tolerance facilitates the evolution of resistance. The reason is that the effect of environmental stressors on stress tolerance has to my knowledge so far only been shown for drug tolerance, not for tolerance to an antagonistic species.

Weaknesses:

I did not detect any weaknesses that would require a major reorganization of the paper, or that may require crucial new experiments without which the paper should not be published. The originally submitted paper needed some work in clarifying specific and central conclusions that the authors draw, which the authors have done during revision.

<https://doi.org/10.7554/eLife.105703.2.sa2>

Reviewer #2 (Public review):

Summary:

An initial screening of pretreatment with different stress treatments of *K. pneumoniae* allowed the identification of heat stress as a protection factor against the infection of the lytic phage Kp11. Then experiments prove that this is mediated not by an increase of phage resistant bacteria but due to an increase in phage transient tolerant population, that the authors identified as bacteriophage persistence in analogy to antibiotic persistence. Then they proved that phage persistence mediated by heat shock enhanced the evolution of bacterial resistance against the phage. The same trait was observed using other lytic phages, their combinations and two clinical strains, as well as *E. coli* and two T phages, hence the phenomenon may be widespread in enterobacteria.

Next, the elucidation of heat induced phage persistence was done, determining that phage adsorption was not affected but phage DNA internalization was impaired by the heat pretreatment, likely to alterations in the bacterial envelope, including the downregulation of envelope proteins and of LPS; furthermore, heat treated bacteria were less sensitive to polymyxins due to the decrease in LPS.

Finally, cyclic exposure to heat stress allowed the isolation of a mutant that was both resistant to heat treatment, polymyxins and lytic phage, that mutant had alterations in PspA protein that allowed a gain of function and that promoted the reduction of capsule production and loss of its structure; nevertheless this mutant was severely impaired in immune evasion as it was easily cleared from mice blood, evidencing the trade-off's between phage/heat and antibiotic resistance and the ability to counteract the immune response.

Strengths:

The experimental design and the sequence in which they are presented is ideal for the understanding of their study and the conclusions are supported by the findings, also the discussion points out the relevance of their work particularly in the effectiveness of phage therapy and allow the design of strategies to improve their effectiveness.

Weaknesses:

In its present form it lacks the incorporation of some relevant previous work that explored the role of heat stress in phage susceptibility, antibiotic susceptibility, trade offs between phage resistance and resistance against other kinds of stress, virulence, etc. and the fact that exposure to lytic phages induces antibiotic persistence.

Comments on revised version:

Thanks for addressing most of my comments; however, although you replied this in the rebuttal:

"Thank you for highlighting these important studies. We have incorporated the work by Majkowska-Skrobek et al. (2021), Gordillo Altamirano et al. (2021), and García-Cruz et al. (2024) into the discussion "

I was not able to find the new section in the discussion of the manuscript.

<https://doi.org/10.7554/eLife.105703.2.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public review):

Summary:

*In this interesting and original paper, the authors examine the effect that heat stress can have on the ability of bacterial cells to evade infection by lytic bacteriophages. Briefly, the authors show that heat stress increases the tolerance of *Klebsiella pneumoniae* to infection by the lytic phage Kp11. They also argue that this increased tolerance facilitates the evolution of genetically encoded resistance to the phage. In addition, they show that heat can reduce the efficacy of phage therapy. Moreover, they define a likely mechanistic reason for both tolerance and genetically encoded resistance. Both lead to a reorganization of the bacterial cell envelope, which reduces the likelihood that phage can successfully inject their DNA.*

Strengths:

I found large parts of this paper well-written and clearly presented. I also found many of the experiments simple yet compelling. For example, the experiments described in Figure 3 clearly show that prior heat exposure can affect the efficacy of phage therapy. In addition, the experiments shown in Figures 4 and 6 clearly demonstrate the likely mechanistic cause of this effect. The conceptual Figure 7 is clear and illustrates the main ideas well. I think this paper would work even without its central claim, namely that tolerance facilitates the evolution of resistance. The reason is that the effect of environmental stressors on stress tolerance has to my knowledge so far only been shown for drug tolerance, not for tolerance to an antagonistic species.

Weaknesses:

I did not detect any weaknesses that would require a major reorganization of the paper, or that may require crucial new experiments. However, the paper needs some work in clarifying specific and central conclusions that the authors draw. More specifically, it needs to improve the connection between what is shown in some figures, how these figures are described in the caption, and how they are discussed in the main text. This is especially glaring with respect to the central claim of the paper from the title, namely that tolerance facilitates the evolution of resistance. I am sympathetic to that claim, especially because this has been shown elsewhere, not for phage resistance but for antibiotic resistance. However, in the description of the results, this is perhaps the weakest aspect of the paper, so I'm a bit mystified as to why the authors focus on this claim. As I mentioned above, the paper could stand on its own even without this claim.

Thank you for your feedback. We understand your concern regarding the central claim that tolerance facilitates the evolution of resistance, while the paper can stand on its own without this claim, we think it provides an important layer to the interpretation of our findings. Considering your comments, we plan to revise the title and adjust to “Heat Stress Induces Phage Tolerance in Bacteria”.

More specific examples where clarification is needed:

(1) A key figure of the paper seems to be Figure 2D, yet it was one of the most confusing figures. This results from a mismatch between the accompanying text starting on line 92 and the figure itself. The first thing that the reader notices in the figure itself is the huge discrepancy between the number of viable colonies in the absence of phage infection at the two-hour time point. Yet this observation is not even mentioned in the main text. The exclusive focus of the main text seems to be on the right-hand side of the figure, labeled "+Phage". It is from this right-hand panel that the authors seem to conclude that heat stress facilitates the evolution of resistance. I find this confusing, because there is no difference between the heat-treated and non-treated cells in survivorship, and it is not clear from this data that survivorship is caused by resistance, not by tolerance/persistence. (The difference between tolerance and resistance has only been shown in the independent experiments of Figure 1B.)

Thank you for your helpful comment. Figure 2d presents colony counts from a plating assay following the phage killing experiment in Figure 2c. Bacteria collected after 0 and 2 hours of phage exposure were plated on both phage-free (-phage) and phage-containing (+phage) plates. The "-phage" condition reflects total survivors, while the "+phage" condition indicates the resistant subset.

As seen in Figure 2d (left part), heat-treated bacteria showed markedly higher survival on phage-free plates than untreated cells, which were largely eliminated by phage. However, resistant colony counts on phage-containing plates were similar between two groups (as shown in figure 2d right part), suggesting that heat stress increased survival but did not promote resistance.

To clarify, we have revised the labels in Figure 2d as follows: "Total" will replace "-phage" to indicate the total survivors from the phage killing assay, and "Resisters" will replace "+phage" to indicate the resistant survivors, which are detected on phage-containing plates. This adjustment should eliminate any confusion and better reflect the experimental design.

Figure 2F supports the resistance claim, but it is not one of the strongest experiments of the paper, because the author simply only used "turbidity" as an indicator of resistance. In addition, the authors performed the experiments described therein at small population sizes to avoid the presence of resistance mutations. But how do we know that the turbidity they describe does not result from persisters?

I see three possibilities to address these issues. First, perhaps this is all a matter of explaining and motivating this particular experiment better. Second, the central claim of the paper may require additional experiments. For example, is it possible to block heat induced tolerance through specific mutations, and show that phage resistance does not evolve as rapidly if tolerance is blocked? A third possibility is to tone down the claim of the paper and make it about heat tolerance rather than the evolution of heat resistance.

Thank you for your thoughtful comment. We appreciate the opportunity to clarify the interpretation of Figure 2f and the rationale behind the experimental design. We agree that turbidity alone cannot fully distinguish resistance from persistence. However, our earlier experiments (Figures 2d and 2e) demonstrated that heat-treated survivors remained largely susceptible to phage, indicating that heat stress does not directly induce resistance. This led us to hypothesize that heat enhances phage tolerance, which in turn increases the likelihood of resistance emergence during subsequent infection.

To test this, we used a low initial bacterial population ($\sim 10^3$ CFU per well) to minimize the chance of pre-existing resistance. Bacteria were exposed to phages at MOIs of 1, 10, and 100 and incubated for 24 hours in 100 μ L volumes. This setup ensured:

- (1) The low initial population minimizes the presence of pre-existing resistant mutants, ensuring that any phage-resistant bacteria observed arise during the infection process.
- (2) The high MOI (≥ 1) ensures that each bacterial cell has a high probability of infection by at least one phage.
- (3) The small volume (100 μ L per well) maximizes the interaction between bacteria and phages, ensuring rapid infection of susceptible bacteria, which leads to clear wells. If resistant mutants arise, they will grow and cause turbidity.

Thus, the turbidity observed in heat-treated samples reflects de novo emergence and outgrowth of resistant mutants from a tolerant population. This assay supports the idea that heat-induced tolerance increases the probability of resistance evolution, rather than directly causing resistance.

We have revised the text to better explain this experimental logic and adjust the framing of our conclusions accordingly.

A minor but general point here is that in Figure 2D and in other figures, the labels "-phage" and "+phage" do not facilitate understanding, because they suggest that cells in the "-phage" treatment have not been exposed to phage at all, but that is not the case. They have survived previous phage treatment and are then replated on media lacking phage.

Thank you for your valuable comment. To clarify, we have revised the labels in Figure 2d as follows: "Total" will replace "-phage" to indicate the total survivors from the phage killing assay, and "Resisters" will replace "+phage" to indicate the resistant survivors, which are detected on phage-containing plates.

*(2) Another figure with a mismatch between text and visual materials is Figure 5, specifically Figures 5B-F. The figure is about two different mutants, and it is not even mentioned in the text how these mutants were identified, for example in different or the same replicate populations. What is more, the two mutants are not discussed at all in the main text. That is, the text, starting on line 221 discusses these experiments as if there was only one mutant. This is especially striking as the two mutants behave very differently, as, for example, in Figure 5C. Implicitly, the text talks about the mutant ending in "...C2", and not the one ending in "...C1". To add to the confusion, the text states that the (C2) mutant shows a change in the *pspA* gene, but in Figure 5f, it is the other (undiscussed) mutant that has a mutation in this gene. Only *pspA* is discussed further, so what about the other mutants? More generally, it is hard to believe that these were the only mutants that occurred in the genome during experimental evolution. It would be useful to give the reader a 2-3 sentence summary of the genetic diversity that experimental evolution generated.*

Thank you for your thoughtful comment. In our heat treatment evolutionary experiment, we isolated six distinct bacterial clones, of which two are highlighted in the manuscript as representative examples. One clone, BC2G11C1, acquired both heat tolerance and phage resistance, while another clone, BC3G11C2, became heat-tolerant but did not develop resistance to phage infection. This variation highlights the inherent diversity in evolutionary responses when exposed to selective pressures. It demonstrates that not all evolutionary pathways lead to the same outcome, even under similar stress conditions. This variability is a key observation in our study, illustrating that different genetic adaptations may arise depending on the specific mutations or genetic context, and not every strain will evolve phage resistance in parallel with heat tolerance. We have updated the manuscript to better reflect this diversity in the evolutionary trajectories observed.

Reviewer #2 (Public review):

Summary:

An initial screening of pretreatment with different stress treatments of K. pneumoniae allowed the identification of heat stress as a protection factor against the infection of the lytic phage Kp11. Then experiments prove that this is mediated not by an increase of phage-resistant bacteria but due to an increase in phage transient tolerant population, which the authors identified as bacteriophage persistence in analogy to antibiotic persistence. Then they proved that phage persistence mediated by heat shock enhanced the evolution of bacterial resistance against the phage. The same trait was observed using other lytic phages, their combinations, and two clinical strains, as well as E. coli and two T phages, hence the phenomenon may be widespread in enterobacteria.

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Weaknesses:

In its present form, it lacks the incorporation of some relevant previous work that explored the role of heat stress in phage susceptibility, antibiotic susceptibility, tradeoffs between phage resistance and resistance against other kinds of stress, virulence, etc., and the fact that exposure to lytic phages induces antibiotic persistence.

Thank you for your insightful comments. I appreciate your suggestion regarding the inclusion of relevant previous works. I have now incorporated additional citations to discuss these points, including studies on the relationship between heat stress and antibiotic resistance, as well as the tradeoffs between phage resistance and other stress factors.

Reviewer #3 (Public review):

PspA, a key regulator in the phage shock protein system, functions as part of the envelope stress response system in bacteria, preventing membrane depolarization and ensuring the envelope stability. This protein has been associated in the Quorum Sensing network and biofilm formation. (Moscoso M., Garcia E., Lopez R. 2006. Biofilm formation by Streptococcus pneumoniae: role of choline, extracellular DNA, and capsular polysaccharide in microbial accretion. J. Bacteriol. 188:7785-7795; Vidal JE, Ludewick HP, Kunkel RM, Zähler D, Klugman KP. The LuxS-dependent quorum-sensing system

regulates early biofilm formation by Streptococcus pneumoniae strain D39. Infect Immun. 2011 Oct;79(10):4050-60.)

It is interesting and very well-developed.

(1) Could the authors develop experiments about the relationship between Quorum Sensing and this protein?

(2) It would be interesting to analyze the link to phage infection and heat stress in relation to Quorum. The authors could study QS regulators or AI2 molecules.

Thank you for your insightful comments and for bringing up the role of PspA in quorum sensing and biofilm formation. However, we would like to clarify a potential misunderstanding: the PspA discussed in our manuscript refers to phage-shock protein A, a key regulator in the bacterial envelope stress response system. This is distinct from the pneumococcal surface protein A, which has been associated with quorum sensing and biofilm formation in *Streptococcus pneumoniae* (as referenced in your comment).

To avoid any confusion for readers, we will ensure that our manuscript explicitly states “phage-shock protein A (PspA)” at its first mention. We appreciate your feedback and hope this clarification addresses your concern.

(3) Include the proteins or genes in a table or figure from lytic phage Kp11 (GenBank: ON148528.1).

Thank you for your helpful suggestion. We have now included a figure, as appropriate summarizing the proteins of the lytic phage Kp11 (GenBank: ON148528.1) in supplementary Figure S1.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Issues unrelated to those discussed in the public review

(1) Figure 4a and its caption describe an evolution experiment, but they do not mention how many cycles of high-temperature treatment and growth this experiment lasted. I assume it lasted for more than one cycle, because the methods section mentions "cycles", but the number is not provided.

Thank you for pointing this out. The evolutionary experiment shown in Figure 5a involved 11 cycles of high-temperature treatment and growth. We have now explicitly stated this in the figure legend to ensure clarity: BC: Batch culture, G: Evolution cycle number, C: Colony. BC2G11C1 refers to the first colony from batvh culture 2 after 11 rounds of heat treatment.

(2) It is not clear what Figure 5F is supposed to show. What are the gray boxes? The caption claims that the figure shows non-synonymous mutations, but the only information it contains is about genes that seem to be affected by mutation. Judging from the mismatch between the main text and the figure, the mutants with these mutations may actually be mislabeled.

Thank you for your careful review. Figure 5f highlights the non-synonymous mutations identified in the evolved strains. The gray boxes represent the ancestral strain's whole genome without mutations, serving as a control. The corresponding labels indicate the specific mutations found in each evolved strain. We have clarified this in the figure caption to

improve clarity. Additionally, we have carefully reviewed the labeling to ensure accuracy and consistency between the figure, main text, and sequencing data.

(3) I think that the acronym NC, which is used in just about every figure, is explained nowhere in the paper. Spell out all acronyms at first use.

Thank you for pointing this out. We have reviewed ensure that NC is clearly defined at its first mention in the text and figure legends to improve clarity. Additionally, we have reviewed the manuscript to ensure that all acronyms are properly introduced when first used.

(4) The same holds for the acronym N.D. This is an especially important oversight because N.D. could mean "not determined" or "not detectable", which would lead to very different interpretations of the same figure.

Thank you for your careful review. We have clarified the meaning of N.D., which stands for non-detectable, at its first use to avoid ambiguity and ensure accurate interpretation in the figure legend. Additionally, we have reviewed the manuscript to ensure that all acronyms are clearly defined.

(5) The panel labels (a,b, etc.) in all figure captions are very difficult to distinguish from the rest of the text, and should be better highlighted, for example by using a bold font. However, this is a matter of journal style and will probably be fixed during typesetting.

Thank you for your suggestion. We have adjusted the figure captions to better distinguish panel labels, such as using bold font, to improve readability and final formatting will follow the journal's style during typesetting.

(6) Line 224: enhanced insusceptibility -> reduced susceptibility.

Thank you for your suggestion. We have revised “enhanced insusceptibility” to “reduced susceptibility” for clarity and precision.

(7) Line 259: mice -> mouse.

Thank you for catching this. We have corrected “mice” to “mouse”.

Reviewer #2 (Recommendations for the authors):

I have no concerns about the experimental design and conclusions of your work; however, I strongly recommend incorporating several relevant pieces of the literature related to your work, in the discussion of your manuscript, specifically:

(1) Previous studies about the role of heat stress in phage infections, see:

Greenrod STE, Cazares D, Johnson S, Hector TE, Stevens EJ, MacLean RC, King KC. Warming alters life-history traits and competition in a phage community. Appl Environ Microbiol. 2024 May 21;90(5):e0028624. doi: 10.1128/aem.00286-24. Epub 2024 Apr 16. PMID: 38624196; PMCID: PMC11107170.

Thank you for your thoughtful comment. We have ensured to incorporate the study by Greenrod et al. (2024) into the discussion to enrich the context of our findings. As this article pointed out, a temperature of 42°C can indeed limit phage infection in bacteria, acting as a barrier from the phage's perspective. Our study builds on this by demonstrating that bacteria pre-treated with high temperatures exhibit tolerance to phage infection. These findings, together with the work you referenced, underscore the importance of heat stress or elevated

temperature in host-phage interactions, with 42°C being particularly relevant in the context of fever. We will make sure to clarify this connection in our revised manuscript.

(2) *The effect of heat stress and the tolerance/resistance against other antibiotics besides polymyxins, see:*

Lv B, Huang X, Lijia C, Ma Y, Bian M, Li Z, Duan J, Zhou F, Yang B, Qie X, Song Y, Wood TK, Fu X. Heat shock potentiates aminoglycosides against gram-negative bacteria by enhancing antibiotic uptake, protein aggregation, and ROS. *Proc Natl Acad Sci U S A*. 2023 Mar 21;120(12):e2217254120. doi: 10.1073/pnas.2217254120. Epub 2023 Mar 14. PMID: 36917671; PMCID: PMC10041086.

Thank you for bringing this study to our attention. We have incorporated the findings from Lv et al. (2023) into the discussion of our manuscript, highlighting how sublethal temperatures may facilitate the killing of bacteria by antibiotics like kanamycin. This is consistent with our data showing enhanced susceptibility of heat-shocked bacteria to kanamycin. The study also provides insights into the potential role of PMF, which is relevant to our work on PspA, and strengthens the broader context of heat stress influencing both antibiotic resistance and tolerance.

(3) *Perhaps the most relevant overlooked fact was that recently it was demonstrated for E. coli, Klebsiella and Pseudomonas that pretreatment with lytic phages induced antibiotic persistence! Please discuss this finding and its implications for your work, see:*

Fernández-García L, Kirigo J, Huelgas-Méndez D, Benedik MJ, Tomás M, García-Contreras R, Wood TK. Phages produce persisters. *Microb Biotechnol*. 2024 Aug;17(8):e14543. doi: 10.1111/1751-7915.14543. PMID: 39096350; PMCID: PMC11297538.

Sanchez-Torres V, Kirigo J, Wood TK. Implications of lytic phage infections inducing persistence. *Curr Opin Microbiol*. 2024 Jun;79:102482. doi: 10.1016/j.mib.2024.102482. Epub 2024 May 6. PMID: 38714140.

Thank you for suggesting this important reference. We agree that the phenomenon of phage-induced bacterial persistence is highly relevant to our study. While our manuscript focuses on the role of heat stress in bacterial tolerance and resistance, we acknowledge that bacterial persistence against phages is an established concept. We have incorporated this finding into our discussion, emphasizing how persistence and tolerance can overlap in their effects on bacterial survival, especially under stress conditions like heat treatment. This will provide a more comprehensive understanding of how phage interactions with bacteria can lead to both persistence and resistance.

(4) *Finally, you observed a tradeoff of the pspA* mutant increased phage/heat/polymyxin resistance and decreased immune evasion (perhaps by being unable to counteract phagocytosis), those tradeoffs between gaining phage resistance but losing resistance to the immune system, virulence impairment and resistance against some antibiotics had been extensively documented, see:*

Majkowska-Skrobek G, Markwitz P, Sosnowska E, Lood C, Lavigne R, Drulis-Kawa Z. The evolutionary trade-offs in phage-resistant *Klebsiella pneumoniae* entail cross-phage sensitization and loss of multidrug resistance. *Environ Microbiol*. 2021 Dec;23(12):7723-7740. doi: 10.1111/1462-2920.15476. Epub 2021 Mar 27. PMID: 33754440.

Gordillo Altamirano F, Forsyth JH, Patwa R, Kostoulas X, Trim M, Subedi D, Archer SK, Morris FC, Oliveira C, Kielty L, Korneev D, O'Bryan MK, Lithgow TJ, Peleg AY, Barr JJ. Bacteriophage-resistant *Acinetobacter baumannii* are resensitized to antimicrobials. *Nat*

Microbiol. 2021 Feb;6(2):157-161. doi: 10.1038/s41564-020-00830-7. Epub 2021 Jan 11. PMID: 33432151.

García-Cruz JC, Rebollar-Juarez X, Limones-Martinez A, Santos-Lopez CS, Toya S, Maeda T, Ceapă CD, Blasco L, Tomás M, Díaz-Velásquez CE, Vaca-Paniagua F, Díaz-Guerrero M, Cazares D, Cazares A, Hernández-Durán M, López-Jácome LE, Franco-Cendejas R, Husain FM, Khan A, Arshad M, Morales-Espinosa R, Fernández-Presas AM, Cadet F, Wood TK, García-Contreras R. Resistance against two lytic phage variants attenuates virulence and antibiotic resistance in *Pseudomonas aeruginosa*. *Front Cell Infect Microbiol.* 2024 Jan 17;13:1280265. doi: 10.3389/fcimb.2023.1280265. Erratum in: *Front Cell Infect Microbiol.* 2024 Mar 06;14:1391783. doi: 10.3389/fcimb.2024.1391783. PMID: 38298921; PMCID: PMC10828002.

Thank you for highlighting these important studies. We have incorporated the work by Majkowska-Skrobek et al. (2021), Gordillo Altamirano et al. (2021), and García-Cruz et al. (2024) into the discussion to provide further context to the evolutionary trade-offs observed in our study. The findings in these studies, which describe the cross-sensitization to antimicrobials and the loss of multidrug resistance in phage-resistant bacteria, align with our observations of trade-offs in the *pspA* mutant. Specifically, our results show that while the *pspA* mutant exhibits increased resistance to phage, heat, and polymyxins, it also experiences a decrease in immune evasion and potential virulence. These trade-offs are significant in understanding the broader consequences of developing resistance to phages and other stressors.

<https://doi.org/10.7554/eLife.105703.2.sa0>